THE NONSEPARABLE CASE OF KADISON'S PROBLEM ON ORTHONORMAL BASES OF UNITARIES FOR TYPE II_1 FACTORS

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ABSTRACT. In 1967, Kadison asked "does every type II_1 factor have an orthonormal (with respect to the trace) basis consisting of unitaries?" In a previous paper [HTZ26], He, Tang, and Zhang resolved Kadison's problem in the separable case. We prove the complementary nonseparable case and thereby resolve Kadison's problem in full. In fact, the basis may be chosen to consist of self-adjoint unitaries. The proof combines a relative norming lemma under small-density constraints, a finite-layer certification scheme ensuring that the relevant Hilbert-space projections are represented by bounded elements of the ambient factor, and a transfinite extension along the density character of $L^2(M,\tau)$.

CONTENTS

1. INTRODUCTION

Let M be a finite von Neumann algebra equipped with a faithful normal tracial state τ. We denote by $L^2(M,\tau)$ the Hilbert space completion of M for the norm $\|x\|_2 = \tau(x^*x)^{1/2}$. For $x \in M$, its image in $L^2(M,\tau)$ will be denoted by $\widehat{x}$. We write $M_{\mathrm{sa}} = \{x \in M : x = x^*\}$ for the self-adjoint part of M. We also denote by $L^2(M,\tau)_{\mathrm{sa}}$ the real Hilbert space obtained as the $\|\cdot\|_2$-closure of M_{sa} in $L^2(M,\tau)$. We use the convention that the inner product is linear in the first variable:

$$\langle \widehat{x}, \widehat{y} \rangle_2 = \tau(y^*x), \qquad x, y \in M.$$

When no confusion is possible, we identify an element $x \in M$ with its vector $\widehat{x} \in L^2(M,\tau)$. Throughout the paper, M acts on $L^2(M,\tau)$ by left multiplication. A vector $\xi \in L^2(M,\tau)$ is called a *trace vector* for this action if

$$\langle x\xi, \xi \rangle_2 = \tau(x), \qquad x \in M.$$

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By an *orthonormal basis* we always mean a Hilbert-space orthonormal basis, namely an orthonormal family whose closed linear span is the whole Hilbert space. In the nonseparable case such a basis may be uncountable. In this paper, "nonseparable" refers to the Hilbert-space density character $\operatorname{dens} L^2(M,\tau)$, not to an arbitrary representation of M.

We also denote by $\mathcal{U}(M)$ the unitary group of M:

$$\mathcal{U}(M) = \{u \in M : u^*u = uu^* = 1\}.$$

A self-adjoint unitary will be called a *symmetry*.

In 1967, at the Baton Rouge conference [Kad67], Kadison asked the following trace-vector orthonormal basis problem for II_1 factors; see also [Ge03, p. 622, Problem 13] or [Pet20, Problem S.2].

Problem 1.1 (Kadison)**.** *Let M be a type II_1 factor with a faithful normal tracial state τ. Does $L^2(M,\tau)$ admit a Hilbert orthonormal basis consisting of trace vectors?*

In the tracial standard representation, Problem 1.1 is also naturally reformulated as a *unitary basis problem.* Indeed, if $u \in \mathcal{U}(M)$, then

$$\langle x\widehat{u}, \widehat{u}\rangle_2 = \tau(u^*xu) = \tau(x), \qquad x \in M,$$

so $\widehat{u}$ is a trace vector. Conversely, as recalled in Proposition 2.1, every trace vector in $L^2(M,\tau)$ is represented by a unitary in M. Thus, Problem 1.1 is equivalent to asking whether $L^2(M,\tau)$ has a Hilbert orthonormal basis consisting of unitaries in M.

The study of the separable case of Problem 1.1 spans nearly 60 years. Early affirmative results were obtained by explicit constructions for group von Neumann algebras associated with countable discrete groups and for finite von Neumann algebras arising from group measure space constructions; see, for example, [Chi73, Cho87]. In 2023, De and Mukherjee [DM23] proved that every separable diffuse finite von Neumann algebra admits a uniformly bounded self-adjoint orthonormal basis in its GNS space. Their construction, however, does not yield unitaries. The separable case of Problem 1.1 was finally settled by He, Tang and Zhang in [HTZ26], where a key ingredient is a noncommutative Lyapunov theorem due to Akemann and Weaver [AW03]. A related application of Lyapunov's convexity theorem appears in [ACFI26, Lemma 2.2], where it is used to construct sequences of unitaries satisfying prescribed trace-orthogonality conditions.

Thus, only the nonseparable case of Problem 1.1 remains open. A natural first attempt is to adapt the method of He, Tang, and Zhang [HTZ26] to the nonseparable setting, and this was indeed our starting point. Their argument, however, is intrinsically separable. The greedy construction proceeds in such a way that, at each finite stage, only finitely many orthogonality constraints have to be imposed, so that the Akemann–Weaver noncommutative Lyapunov theorem can be applied. A transfinite version of this construction breaks down already at the first limit stage: one would need to produce a symmetry satisfying infinitely many simultaneous orthogonality constraints, which is not covered by the finite form of the Lyapunov theorem underlying their argument. Nevertheless, the ideas and techniques developed in [HTZ26] continue to play an important role in our study of the nonseparable case, for example in Lemmas 3.4, 6.1, and 6.3.

In this paper, we prove a slightly stronger statement than what is required in Problem 1.1: the orthonormal basis may be chosen to consist entirely of self-adjoint unitaries. More precisely, we prove the following.

Theorem 1.2. *Let M be a type II_1 factor with a faithful normal tracial state τ, and set $\kappa = \operatorname{dens} L^2(M,\tau)$. If $\kappa > \aleph_0$, then there is a family $(s_i)_{i\in I}$ of self-adjoint unitaries in M, indexed by a set I with $|I| = \kappa$, such that $\{\widehat{s_i} : i \in I\}$ is a Hilbert-space orthonormal basis of $L^2(M,\tau)$.*

As an immediate consequence of Theorem 1.2, we obtain the following.

Corollary 1.3. *Let M be a nonseparable type* II_1 *factor with a faithful normal tracial state τ. Then $L^2(M,\tau)$ admits a Hilbert-space orthonormal basis consisting of self-adjoint unitaries in M, and hence of trace vectors. Together with the separable case proved by He–Tang–Zhang* [HTZ26]*, this settles Problem 1.1 in the affirmative.*

Theorem 1.2 requires a different mechanism. One must extend bases through subalgebras of smaller density character while controlling orthogonality to previously chosen, possibly very large, backgrounds. The finite-layer certified background formalism records enough von Neumann algebraic structure to ensure that the relevant Hilbert-space projections remain bounded elements of M. This allows a transfinite extension argument over the density character of $L^2(M,\tau)$.

Remark 1.4. Theorem 1.2 removes the separability assumption from [HTZ26, Theorem 1] within the class of type II_1 factors. Our argument does not directly yield the corresponding nonseparable statement for arbitrary diffuse finite von Neumann algebras.

Indeed, the proof of Lemma 3.4 relies on the factor property through Lemma 3.2: if $\kappa = \operatorname{dens} L^2(M,\tau)$, then every nonzero corner eMe has L^2-density κ. This ensures that fewer than κ constraints cannot exhaust the self-adjoint L^2-space of a nonzero spectral corner, which is the key point in Lemma 3.3.

For a finite von Neumann algebra with nontrivial center, nonzero corners may have strictly smaller density. For instance, if $M = M_0 \oplus M_1$, with M_0 separable diffuse and $\operatorname{dens} L^2(M_1) = \kappa > \aleph_0$, then $\operatorname{dens} L^2(M) = \kappa$, while for the central projection $e = (1,0)$, $\operatorname{dens} L^2(eMe) = \aleph_0 < \kappa$. Hence a constraint family of cardinality $< \kappa$ need not be small relative to the corner in which the perturbation is required. Accordingly, the present density-gap argument cannot be applied verbatim to the nonseparable analogue of [HTZ26, Theorem 1] in the general diffuse finite case. A center-wise refinement, or some other replacement for the uniform corner-density argument, would be needed.

We introduce the notain will be used in this paper. If $N \subset M$ is a von Neumann subalgebra, we denote by $E_N : M \to N$ the unique τ-preserving normal conditional expectation. Its L^2-extension is the orthogonal projection onto $L^2(N,\tau)$. In particular,

$$E_N(x) = 0 \quad \iff \quad x \perp L^2(N,\tau), \qquad x \in M. \tag{1.1}$$

If $F \subset M$, the notation $E_N(F) = 0$ means that $E_N(f) = 0$ for every $f \in F$. If $F, G \subset L^2(M,\tau)$, or if one of them is a closed subspace, then $F \perp G$ means that every element of F is orthogonal to every element of G. We write $F \dot\cup G$ for a disjoint union and $\mathcal{P}(F)$ for the power set of F. If $\mathcal{P}_0 \subset M$, then $*\text{-alg}_{\mathbb{Q}+i\mathbb{Q}}(\mathcal{P}_0)$ denotes the unital $*$-algebra over $\mathbb{Q}+i\mathbb{Q}$ generated by $\mathcal{P}_0$.

Sketch of the proof and organization of the paper. The proof of Theorem 1.2 consists of four main steps.

Step 1: Trace-zero reduction and relative norming. For a finite von Neumann algebra P, put

$$H_0(P) = L^2(P,\tau)_{\mathrm{sa}} \ominus \mathbb{R}1.$$

By Proposition 2.1 and Lemma 2.2, it is enough to construct an orthonormal basis of $H_0(M)$ consisting of trace-zero symmetries.

The key analytic input is Lemma 3.4. Suppose that $N \subset M$ has L^2-density strictly smaller than that of M, and that $0 \neq r = r^* \in M$ is orthogonal both to $L^2(N,\tau)_{\mathrm{sa}}$ and to fewer than $\operatorname{dens} L^2(M,\tau)$ additional self-adjoint constraints. The lemma produces $s \in \mathcal{S}_0(M)$ satisfying the same orthogonality conditions and

$$\langle r, s\rangle_2 = \frac{\|r\|_2^2}{\|r\|_\infty}. \tag{1.2}$$

Its proof combines the corner perturbation lemma, Lemma 3.3, with a Krein–Milman argument. The strict density gap is used only in this step.

Step 2: Bounded projections through certified backgrounds. For an uncountable orthonormal family, the Hilbert-space projection onto an arbitrary subfamily need not be represented by an element of M. To retain bounded representatives, Sections 4 and 5 introduce completed blocks and finite-layer certificates.

A completed block has the form

$$L^2(Q,\tau)_{\mathrm{sa}} = L^2(A,\tau)_{\mathrm{sa}} \oplus \overline{\mathrm{span}_{\mathbb{R}}}T,$$

and Lemma 4.2 gives the bounded projection formula

$$P_T^{\mathfrak{B}}(x) = E_Q(x) - E_A(E_Q(x)), \qquad x = x^* \in M.$$

Finite-layer certificates are obtained by adjoining completed blocks one layer at a time. Proposition 5.4 shows that every small subfamily of a certified background can be enlarged to a still small certified subbackground. Consequently, all projections used in the subsequent construction are represented by bounded self-adjoint elements of M.

Step 3: Countable and all-cardinal extension. Lemma 6.3 is the basic construction step. Given a von Neumann subalgebra N of smaller density, a certified background $\Omega \perp L^2(N,\tau)_{\mathrm{sa}}$, and countable sets of elements to be absorbed, it constructs a separable completed block

$$L^2(R,\tau)_{\mathrm{sa}} = L^2(C,\tau)_{\mathrm{sa}} \oplus \overline{\mathrm{span}_{\mathbb{R}}}(\Omega_0 \cup U),$$

where $C \subset N$, the set $\Omega_0 \subset \Omega$ is closed and certified, and U is a countable orthonormal family of trace-zero symmetries orthogonal to both N and Ω.

For every eventual self-adjoint $*$-polynomial word y, one considers the bounded residual

$$b(y) = y - E_N(y) - P_\Omega(y). \tag{1.3}$$

Whenever its component outside the span of the previously chosen symmetries is nonzero, the norming identity (1.2) supplies a new symmetry and decreases the squared distance quantitatively. Lemma 6.1 schedules each eventual word at times n_k satisfying

$$\sum_k \frac{1}{n_k} = \infty.$$

The sparse greedy descent argument then forces every residual (1.3) into $\overline{\mathrm{span}_{\mathbb{R}}}U$.

Proposition 7.3 upgrades the countable construction to every cardinal

$$\aleph_0 \le \lambda < \operatorname{dens} L^2(M,\tau)$$

by transfinite induction. No regularity assumption is imposed on the ambient density character.

Step 4: Relative basis extension and transfinite completion. Proposition 8.1 converts the certified extension theorem into a relative basis-extension principle. If $H_0(N)$ already has a symmetry basis and $X \subset M_{\mathrm{sa}}$ has controlled cardinality, then there is a larger von Neumann algebra $P \supset N \cup X$ and an orthonormal family $U \subset \mathcal{S}_0(P)$ such that

$$H_0(P) = H_0(N) \oplus \overline{\mathrm{span}_{\mathbb{R}}}U.$$

Starting from the separable diffuse abelian algebra of Lemma 8.2, equipped with its Walsh symmetry basis, we apply this extension principle along an L^2-dense family

$$(a_\alpha)_{\alpha<\kappa} \subset H_0(M), \qquad \kappa = \operatorname{dens} L^2(M,\tau).$$

The resulting nested von Neumann algebras N_α have density strictly smaller than κ, and their nested symmetry bases absorb a_α at stage $\alpha + 1$. Their union is therefore an orthonormal basis of $H_0(M)$. Lemma 2.2 then completes the proof of Theorem 1.2.

The organization of the paper follows these four steps. Section 2 contains the trace-vector and trace-zero reductions. Section 3 proves the perturbation and relative norming lemmas. Sections 4 and 5 develop certified backgrounds and their strong closure. Section 6 proves the countable extension theorem, and Section 7 passes to arbitrary cardinals below the ambient density. Finally, Section 8 derives the relative extension property and completes the transfinite recursion.

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AI tool disclosure. ChatGPT 5.6 Sol was used for English-language editing, proofreading, and grammatical corrections, and as an exploratory tool to discuss possible approaches to selected results in this paper, including, for example, Lemma 5.3 and Proposition 5.4. The final mathematical arguments and proofs were independently developed, checked, and written by the human authors.

2. Preliminaries and reductions

For a finite von Neumann algebra (P, τ), we shall write

$$H_0(P) = L^2(P, \tau)_{\mathrm{sa}} \ominus \mathbb{R}1 = \{\xi \in L^2(P, \tau)_{\mathrm{sa}} : \langle \xi, 1 \rangle_2 = 0\}.$$

Thus, if $x = x^* \in P$, then $\widehat{x} \in H_0(P)$ exactly when $\tau(x) = 0$. We also write

$$\mathcal{S}_0(P) = \{s \in P : s = s^* = s^{-1},\ \tau(s) = 0\}$$

for the set of trace-zero symmetries in P. When P is a von Neumann subalgebra of a fixed finite von Neumann algebra, these notations are always understood with respect to the restricted trace.

We first record the following standard reduction from trace vectors to unitaries, which is an immediate consequence of the classical reflection theorem for trace vectors [KR97b, Theorem 7.2.15].

Proposition 2.1. *Let (M, τ) be a finite von Neumann algebra in its tracial standard representation on $L^2(M, \tau)$. A vector $\xi \in L^2(M, \tau)$ is a trace vector for M if and only if $\xi = \widehat{u}$ for some unitary $u \in \mathcal{U}(M)$.*

Proof. If $u \in \mathcal{U}(M)$, then for every $x \in M$,

$$\langle x\widehat{u}, \widehat{u} \rangle_2 = \tau(u^* x u) = \tau(x),$$

so $\widehat{u}$ is a trace vector.

Conversely, suppose that ξ is a trace vector, and let $\Omega = \widehat{1}$. Since Ω is a generating trace vector for M, the classical reflection theorem [KR97b, Theorem 7.2.15] yields a $*$-anti-isomorphism $\Theta : M \longrightarrow M'$ characterized by

$$\Theta(a)\Omega = a\Omega, \qquad a \in M,$$

and, moreover, M' is finite.

Define

$$V_0 : M\Omega \longrightarrow L^2(M, \tau), \qquad V_0(x\Omega) = x\xi.$$

Since ξ is a trace vector, for every $x \in M$,

$$\|V_0(x\Omega)\|_2^2 = \langle x^* x \xi, \xi \rangle_2 = \tau(x^* x) = \|x\Omega\|_2^2.$$

Thus V_0 extends to an isometry $V \in B(L^2(M, \tau))$. Moreover, for $a, x \in M$,

$$V(ax\Omega) = ax\xi = aV(x\Omega),$$

and hence $V \in M'$. Since M' is finite, every isometry in M' is unitary. Therefore $V \in \mathcal{U}(M')$.

By the surjectivity of Θ, there exists $u \in M$ such that $V = \Theta(u)$. Since Θ is a $*$-anti-isomorphism and V is unitary, $u \in \mathcal{U}(M)$. Finally,

$$\xi = V\Omega = \Theta(u)\Omega = u\Omega = \widehat{u}.$$

This proves the proposition. $\square$

The next lemma gives the basic trace-zero reduction needed for the nonseparable case. The same underlying idea also appears in [HTZ26].

Lemma 2.2. *Let M be a II_1 factor. If $B \subset \mathcal{S}_0(M)$ is a Hilbert orthonormal basis of the real Hilbert space $H_0(M)$, then $\{\widehat{1}\} \cup \{\widehat{s} : s \in B\}$ is a Hilbert orthonormal basis of the complex Hilbert space $L^2(M, \tau)$. Moreover, every vector in $\{\widehat{1}\} \cup \{\widehat{s} : s \in B\}$ is a trace vector for the left action of M.*

Proof. Clearly, for self-adjoint $a, b \in M$, $\tau(ba)$ is real. Thus real orthogonality inside $L^2(M, \tau)_{\mathrm{sa}}$ is the same as complex Hilbert-space orthogonality for these vectors. Since B is an orthonormal basis of $H_0(M)$ and $B \subset \mathcal{S}_0(M)$, the family $\{1\} \cup B$ is orthonormal in $L^2(M, \tau)$.

Let $\mathcal{L}$ be the closed complex linear span of $\{1\} \cup B$. Since

$$L^2(M, \tau)_{\mathrm{sa}} = \mathbb{R}1 \oplus H_0(M) = \overline{\mathrm{span}_{\mathbb{R}}}\big(\{1\} \cup B\big),$$

we have $L^2(M, \tau)_{\mathrm{sa}} \subset \mathcal{L}$. Hence $M_{\mathrm{sa}} \subset \mathcal{L}$, and therefore

$$M = M_{\mathrm{sa}} + iM_{\mathrm{sa}} \subset \mathcal{L}.$$

Since M is dense in $L^2(M, \tau)$, it follows that $\mathcal{L} = L^2(M, \tau)$. Thus $\{1\} \cup B$ is an orthonormal basis of the complex Hilbert space $L^2(M, \tau)$.

If $s = s^* = s^{-1} \in M$, then for every $x \in M$,

$$\langle xs, s\rangle_2 = \tau(sxs) = \tau(xss) = \tau(x).$$

Thus every self-adjoint unitary is a trace vector, and so is 1. $\square$

3. A relative norming lemma under small-density constraints

Throughout Sections 3–8, we assume that M is a type II_1 factor equipped with a faithful normal tracial state τ, and we set

$$\kappa = \operatorname{dens} L^2(M, \tau) > \aleph_0.$$

All von Neumann subalgebras are understood to be unital, except that corners such as eMe are taken with their natural unit e. We use density character in the Hilbert norm: $\operatorname{dens} L^2(M)$ is the least cardinality of an L^2-dense subset.

The next lemma records elementary L^2-closure and cardinal facts. More precisely, parts (i) and (ii) are consequences of Kaplansky's density theorem [Tak02, p. 82, Theorem II.4.8] in the tracial GNS representation, part (iii) is the L^2-implementation of the trace-preserving conditional expectation, and part (iv) is a formal consequence of part (iii).

Lemma 3.1. *Let P be a finite von Neumann algebra equipped with a faithful normal tracial state τ.*

- *(i) Let $A_0 \subset P$ be a unital C^*-subalgebra and let $A = A_0'' = W^*(A_0)$. If A_0 has norm density at most μ, then $L^2(A, \tau)$ has density at most $\max\{\mu, \aleph_0\}$.*
- *(ii) Let $(P_i)_{i \in I}$ be a directed increasing family of von Neumann subalgebras of P, and suppose that $P_0 = W^*\Big(\bigcup_{i \in I} P_i\Big)$. Then $L^2(P_0, \tau) = \overline{\bigcup_{i \in I} L^2(P_i, \tau)}^{\|\cdot\|_2}$.*
- *(iii) Let $Q \subset P$ be a von Neumann subalgebra. If $x \in P$ satisfies $\widehat{x} \in L^2(Q, \tau)$, then $x \in Q$.*
- *(iv) If $Y \subset P_{\mathrm{sa}}$ is L^2-dense in $L^2(P, \tau)_{\mathrm{sa}}$, then $W^*(Y) = P$.*

Proof. We regard P in its faithful tracial GNS representation on $L^2(P,\tau)$, acting by left multiplication. Since this representation is faithful and normal, it preserves the von Neumann algebras generated by the subalgebras under consideration.

For (i), put $\kappa = \max\{\mu, \aleph_0\}$. The unit ball $(A_0)_1$ has a norm-dense subset $D \subset (A_0)_1$ with $|D| \le \kappa$. By Kaplansky's density theorem [Tak02, p. 82, Theorem II.4.8], the unit ball of A_0 is strong-$*$ dense in the unit ball of A. Since norm convergence implies strong-$*$ convergence, D is itself strong-$*$ dense in the unit ball of A. Thus, for every $a \in A$ with $\|a\| \le 1$, there is a net $(d_\alpha)_\alpha$ in D such that $d_\alpha \to a$ strongly. In the tracial GNS representation,

$$\|d_\alpha - a\|_2 = \|(d_\alpha - a)\widehat{1}\|_{L^2(P,\tau)} \longrightarrow 0.$$

It follows that D is L^2-dense in the unit ball of A. Consequently, $\bigcup_{n\ge 1} nD$ is L^2-dense in A, and hence in $L^2(A,\tau)$. Since this set has cardinal at most κ, the desired density estimate follows.

For (ii), set $B = \overline{\bigcup_{i\in I} P_i}^{\|\cdot\|}$. Directedness ensures that $\bigcup_i P_i$ is a unital $*$-subalgebra, so B is a C^*-algebra, and $B'' = P_0$. Put $\mathcal{H} = \overline{\bigcup_{i\in I} L^2(P_i,\tau)}^{\|\cdot\|_2} \subset L^2(P_0,\tau)$. Since $\bigcup_i P_i$ is norm dense in B, and since $\|x\|_2 \le \|x\|$, we have $\widehat{B} \subset \mathcal{H}$. By Kaplansky's density theorem [Tak02, p. 82, Theorem II.4.8], the unit ball of B is strongly dense in the unit ball of P_0. Strong convergence in the tracial GNS representation implies convergence on the vector $\widehat{1}$, and hence L^2-convergence. Therefore $\widehat{B}$ is L^2-dense in $L^2(P_0,\tau)$. Since $\mathcal{H}$ is closed and contains $\widehat{B}$, it follows that

$$L^2(P_0,\tau) \subset \mathcal{H}.$$

The reverse inclusion is immediate, and hence

$$L^2(P_0,\tau) = \overline{\bigcup_{i\in I} L^2(P_i,\tau)}^{\|\cdot\|_2}.$$

For (iii), let $E_Q : P \to Q$ be the unique τ-preserving normal conditional expectation. Its existence follows from the finite tracial case of Takesaki's conditional-expectation theorem [Tak72, Section 3, p. 309]; see also [Tak02, Proposition V.2.36]. For $x \in P$ and $q \in Q$, trace preservation and Q-bimodularity give

$$\begin{aligned}\left\langle \widehat{x - E_Q(x)}, \widehat{q} \right\rangle &= \tau\big(q^*(x - E_Q(x))\big)\\ &= \tau\big(E_Q\big(q^*(x - E_Q(x))\big)\big)\\ &= 0.\end{aligned}$$

Hence $e_Q\widehat{x} = \widehat{E_Q(x)}$, where $e_Q : L^2(P,\tau) \to L^2(Q,\tau)$ denotes the orthogonal projection. If $\widehat{x} \in L^2(Q,\tau)$, then $\widehat{x} = e_Q\widehat{x} = \widehat{E_Q(x)}$. The faithfulness of τ therefore yields $x = E_Q(x) \in Q$.

Finally, (iv) is an immediate consequence of (iii). Put $Q = W^*(Y)$. The space $L^2(Q,\tau)_{\mathrm{sa}}$ is a closed real subspace of $L^2(P,\tau)_{\mathrm{sa}}$ containing $\{\widehat{y} : y \in Y\}$. Since Y is L^2-dense in $L^2(P,\tau)_{\mathrm{sa}}$, it follows that $L^2(Q,\tau)_{\mathrm{sa}} = L^2(P,\tau)_{\mathrm{sa}}$. Thus, for every $x \in P_{\mathrm{sa}}$, one has $\widehat{x} \in L^2(Q,\tau)$, and part (iii) gives $x \in Q$. Hence $P_{\mathrm{sa}} \subset Q$, and therefore $P = Q = W^*(Y)$. □

We will only need the following form of the corner-density statement. A stronger version, with no restriction on the density character, is proved in Appendix A.

Lemma 3.2. *Let M be a II_1 factor with $\operatorname{dens} L^2(M) = \kappa > \aleph_0$. If $0 \ne e \in M$ is a projection, then*

$$\operatorname{dens} L^2(eMe) = \kappa,$$

where eMe is equipped with its finite trace inherited from τ.

Proof. Clearly, $\operatorname{dens} L^2(eMe) \le \kappa$. Let $\mu = \operatorname{dens} L^2(eMe)$. Choose $n \in \mathbb{N}$ with $1/n \le \tau(e)$. In a II_1 factor there are orthogonal projections $e_1, \dots, e_n$ with $\sum_i e_i = 1$ and $\tau(e_i) = 1/n$. There are also projections $f_i \le e$ with $\tau(f_i) = 1/n$; these f_i need not be mutually orthogonal. Let v_i be partial isometries with $v_i^* v_i = f_i$ and $v_i v_i^* = e_i$. For $x \in M$,

$$e_i x e_j = v_i \left(v_i^* x v_j\right) v_j^*, \qquad v_i^* x v_j \in f_i M f_j \subset eMe.$$

Thus M is contained in the finite linear span of the subspaces $v_i(eMe)v_j^*$. Each of these subspaces has L^2-density at most μ, so

$$\kappa = \operatorname{dens} L^2(M) \le \max\{\mu, \aleph_0\}.$$

Since $\kappa > \aleph_0$, this inequality forces $\mu > \aleph_0$ and hence $\max\{\mu, \aleph_0\} = \mu$. Therefore $\kappa \le \mu$, and the reverse inequality already observed gives $\mu = \kappa$. □

Next, we establish the following small-constraint perturbation lemma.

Lemma 3.3. *Let M be a II_1 factor with $\operatorname{dens} L^2(M) = \kappa > \aleph_0$ and $\Gamma \subset M_{\mathrm{sa}}$ satisfy $|\Gamma| < \kappa$. If $0 \ne e \in M$ is a projection, then there exists a nonzero $h = h^* \in eMe$ such that*

$$\tau(\gamma h) = 0, \qquad \gamma \in \Gamma.$$

Proof. Let $D = W^*(\{e\} \cup \Gamma)$. If $\lambda = |\Gamma|$, then the unital C^*-algebra generated by $\{e\} \cup \Gamma$ has norm density at most $\max\{\lambda, \aleph_0\} < \kappa$, because rational-coefficient $*$-polynomials in the generators are norm dense in it. By Lemma 3.1(i),

$$\operatorname{dens} L^2(D) < \kappa.$$

Hence also $\operatorname{dens} L^2(eDe) < \kappa$.

By Lemma 3.2,

$$\operatorname{dens} L^2(eMe) = \kappa.$$

Since $L^2(eMe) = L^2(eMe)_{\mathrm{sa}} + iL^2(eMe)_{\mathrm{sa}}$ and similarly for eDe, the real Hilbert space $L^2(eMe)_{\mathrm{sa}}$ has density κ, while $L^2(eDe)_{\mathrm{sa}}$ has density $< \kappa$. Thus $L^2(eDe)_{\mathrm{sa}}$ is a proper closed real subspace of $L^2(eMe)_{\mathrm{sa}}$. Choose $\eta = \eta^* \in L^2(eMe)_{\mathrm{sa}} \setminus L^2(eDe)_{\mathrm{sa}}$. Since $(eMe)_{\mathrm{sa}}$ is L^2-dense in $L^2(eMe)_{\mathrm{sa}}$ and $L^2(eDe)_{\mathrm{sa}}$ is closed, we may choose $a = a^* \in eMe$ so close to η in L^2-norm that $\widehat{a} \notin L^2(eDe)_{\mathrm{sa}}$. In particular, $a \in eMe \setminus eDe$. Let $h = a - E_D(a)$, where $E_D : M \to D$ is the trace-preserving conditional expectation. Since $e \in D$ and $a = eae$, bimodularity of E_D gives

$$E_D(a) = E_D(eae) = eE_D(a)e \in eDe.$$

Thus $h \in eMe$. Moreover $h \ne 0$, because otherwise $a = E_D(a) \in eDe$.

Finally, for $\gamma \in \Gamma \subset D$, trace preservation and D-bimodularity of E_D give

$$\tau(\gamma a) = \tau(E_D(\gamma a)) = \tau(\gamma E_D(a)).$$

Therefore $\tau(\gamma h) = 0$ for every $\gamma \in \Gamma$. □

Lemma 3.4 is the cardinal-density counterpart of the finite-dimensional norming lemma used in the separable proof of Kadison's problem [HTZ26, Lemma 2]. The latter is obtained from the finite-constraint noncommutative Lyapunov theorem of Akemann–Weaver [AW03, Theorem 3.1], which belongs to the operator-algebraic Lyapunov theory developed in [AA91] and is closely connected with the facial geometry of positive contractions studied in [AP92].

There is, however, an essential difference. In Lemma 3.4, the orthogonality conditions consist of all constraints coming from a von Neumann subalgebra N of smaller L^2-density, together with an additional family Γ of cardinality strictly smaller than $\kappa = \operatorname{dens} L^2(M)$. Thus the corresponding affine slice may have infinite, and even uncountable, codimension, so [AW03, Theorem 3.1] is not directly applicable. The new point is that the strict density gap produces a nonzero perturbation in every nontrivial spectral corner while annihilating all the prescribed constraints. This forces an extreme point of the affine slice to be a projection.

Lemma 3.4. *Let M be a* II_1 *factor with* $\operatorname{dens} L^2(M) = \kappa > \aleph_0$. *Let $N \subseteq M$ be a von Neumann subalgebra satisfying* $\operatorname{dens} L^2(N) < \kappa$, *and let* $\Gamma \subseteq M_{\mathrm{sa}}$ *be a set of cardinality less than* κ. *Suppose that* $0 \neq r = r^* \in M$ *satisfies* $E_N(r) = 0$ *and* $\tau(\gamma r) = 0$ *for every* $\gamma \in \Gamma$. *Then there exists* $s \in \mathcal{S}_0(M)$ *such that*

$$E_N(s) = 0, \qquad \tau(\gamma s) = 0 \quad (\gamma \in \Gamma), \qquad \textit{and} \qquad \langle r, s\rangle_2 = \|r\|_2^2/\|r\|_\infty.$$

Proof. Put $c = \|r\|_\infty$ and $q = (1 + r/c)/2$. Then $0 \le q \le 1$. Choose an L^2-dense subset $\mathcal{D} \subset N_{\mathrm{sa}}$ consisting of bounded operators such that

$$|\mathcal{D}| \le \max\{\operatorname{dens} L^2(N), \aleph_0\} < \kappa.$$

Consider the set

$$K = \left\{ 0 \le x = x^* \le 1 : \begin{array}{ll} \tau(yx) = \tau(yq) & (y \in \mathcal{D}), \\ \tau(\gamma x) = \tau(\gamma q) & (\gamma \in \Gamma), \\ \tau(rx) = \tau(rq) & \end{array} \right\}.$$

K is nonempty, because $q \in K$. It is ultraweakly compact and convex: indeed, it is the intersection of the ultraweakly compact positive unit ball of M with ultraweakly closed affine hyperplanes defined by normal functionals. By the Krein–Milman theorem [Con90, p. 142, Theorem V.7.4], the set K has an extreme point; fix one and denote it by p.

We claim that p is a projection. Suppose not. Since $0 \le p \le 1$ and p is not a projection, the spectral projection of p corresponding to the open interval $(0, 1)$ is nonzero. Since $(0, 1) = \bigcup_{n=2}^\infty [1/n, 1 - 1/n]$, there is some $0 < \varepsilon < 1/2$ such that $e = 1_{[\varepsilon, 1-\varepsilon]}(p)$ is nonzero.

Apply Lemma 3.3 to the set $\mathcal{D} \cup \Gamma \cup \{r\}$ and to the projection e. This set has cardinal $< \kappa$. We get a nonzero $h = h^* \in eMe$ such that

$$\tau(yh) = 0 \quad (y \in \mathcal{D}), \qquad \tau(\gamma h) = 0 \quad (\gamma \in \Gamma), \qquad \tau(rh) = 0.$$

After replacing h by a sufficiently small nonzero real scalar multiple, we may assume $\|h\|_\infty < \varepsilon$. Since e is a spectral projection of p, it commutes with p. Also $h = ehe$. Hence, with respect to the decomposition $eL^2(M) \oplus (1 - e)L^2(M)$, the operators $p \pm h$ are block diagonal:

$$p \pm h = (pe \pm h) \oplus p(1 - e).$$

On $eL^2(M)$ we have

$$\varepsilon e \le pe \le (1 - \varepsilon)e.$$

Since $h = h^* \in eMe$, we have

$$-\|h\|_\infty e \le h \le \|h\|_\infty e.$$

As $\|h\|_\infty < \varepsilon$, it follows that

$$0 \le pe \pm h \le e.$$

On $(1 - e)L^2(M)$, the operator $p(1 - e)$ is a positive contraction. Therefore

$$0 \le p \pm h \le 1.$$

The displayed orthogonality relations show that $p+h$ and $p-h$ satisfy all defining affine constraints of K. Thus $p \pm h \in K$, and

$$p = \frac{1}{2}(p + h) + \frac{1}{2}(p - h)$$

is a nontrivial convex decomposition in K, contradicting the extremality of p. Hence p is a projection.

Put $s = 2p - 1$. For $y \in \mathcal{D}$,

$$\tau(yq) = \frac{1}{2}\tau(y) + \frac{1}{2c}\tau(yr) = \frac{1}{2}\tau(y),$$

since $E_N(r) = 0$. Thus $\tau(ys) = 0$ for every $y \in \mathcal{D}$. If $z \in N_{\mathrm{sa}}$, choose $y_i \in \mathcal{D}$ with $\|y_i - z\|_2 \to 0$. Since $s \in M$, the functional $w \longmapsto \tau(ws)$ is L^2-continuous. Hence $\tau(zs) = 0$ for every $z \in N_{\mathrm{sa}}$, and therefore

$$s \perp L^2(N, \tau)_{\mathrm{sa}}.$$

Since $E_N(s)$ is the L^2-orthogonal projection of s onto $L^2(N, \tau)_{\mathrm{sa}}$, it follows that

$$E_N(s) = 0.$$

Similarly, for $\gamma \in \Gamma$,

$$\tau(\gamma q) = \frac{1}{2}\tau(\gamma) + \frac{1}{2c}\tau(\gamma r) = \frac{1}{2}\tau(\gamma),$$

so $\tau(\gamma s) = 0$. Since $1 \in N$, the identity $E_N(s) = 0$ also gives $\tau(s) = 0$. Therefore $s \in \mathcal{S}_0(M)$.

Finally, using the convention that the inner product is linear in the first variable and using traciality,

$$\begin{aligned}
\langle r, s\rangle_2 &= \tau(sr) = \tau(rs) \\
&= \tau(r(2p - 1)) \\
&= 2\tau(rp) - \tau(r) \\
&= 2\tau(rq) - \tau(r) \\
&= 2\left(\frac{1}{2}\tau(r) + \frac{1}{2c}\tau(r^2)\right) - \tau(r) = \frac{\|r\|_2^2}{\|r\|_\infty}.
\end{aligned}$$

□

4. Completed blocks and certified backgrounds

By (1.1), orthogonality relative to a von Neumann subalgebra is naturally expressed through the trace-preserving conditional expectation. This point of view is closely related to Popa's theory of orthogonal subalgebras [Pop83]. There is also a conceptual connection with Pimsner–Popa bases [PP86] and with the bounded homogeneous orthonormal bases of Ioana–Peterson–Popa [IPP08, Definition 4.1], since all of these notions use bounded elements of a von Neumann algebra to represent complete orthogonal systems.

The notion used here is nevertheless different from those module-theoretic bases. We impose scalar L^2-orthogonality rather than orthogonality for a subalgebra-valued inner product, we do not assume finite index, and our basis elements are required to be trace-zero symmetries. The terminology below is introduced in order to record precisely those Hilbert-space decompositions whose orthogonal projections can still be evaluated by bounded elements of the ambient algebra.

Definition 4.1 (Completed block)**.** A *completed block* in M is a triple $\mathfrak{B} = (A, Q, T)$, where $A \subset Q \subset M$ are von Neumann subalgebras and $T \subset \mathcal{S}_0(M)$ is an orthonormal family such that

$$L^2(Q)_{\mathrm{sa}} = L^2(A)_{\mathrm{sa}} \oplus \overline{\mathrm{span}_{\mathbb{R}}} T.$$

The sum is an orthogonal direct sum of real Hilbert spaces.

If a von Neumann subalgebra $N \subset M$ is specified, we say that $\mathfrak{B}$ is a *completed block over* N if, in addition,

$$A \subset N \qquad \text{and} \qquad E_N(t) = 0 \quad (t \in T).$$

In that case $T \perp L^2(N)_{\mathrm{sa}}$, and in particular $T \perp L^2(A)_{\mathrm{sa}}$.

In either case, the displayed decomposition implies $T \subset Q$: indeed $t \in T$ is an element of M whose L^2-vector belongs to $L^2(Q)$, so Lemma 3.1(iii) gives $t \in Q$.

Although the following identity is elementary, it is the basic boundedness mechanism used throughout the certificate construction. It expresses the orthogonal projection onto the complement of a nested subalgebra by trace-preserving conditional expectations, and therefore keeps that projection inside the ambient von Neumann algebra.

Lemma 4.2 (Projection formula)**.** *If $\mathfrak{B} = (A, Q, T)$ is a completed block, then for every $x = x^* \in M$ the orthogonal projection of x onto $\overline{\operatorname{span}_{\mathbb{R}}} T$ is*

$$P_T^{\mathfrak{B}}(x) = E_Q(x) - E_A(E_Q(x)). \tag{4.1}$$

In particular $P_T^{\mathfrak{B}}(x) \in M_{\mathrm{sa}}$.

Proof. The map E_Q is the L^2-orthogonal projection onto $L^2(Q)$ and E_A is the L^2-orthogonal projection from $L^2(Q)$ onto $L^2(A)$. The completed-block decomposition identifies $L^2(Q)_{\mathrm{sa}} \ominus L^2(A)_{\mathrm{sa}}$ with $\overline{\operatorname{span}_{\mathbb{R}}} T$. □

For an arbitrary, possibly uncountable, orthonormal family $\Omega \subset L^2(M)_{\mathrm{sa}}$, the Hilbert-space projection onto $\overline{\operatorname{span}_{\mathbb{R}}}\Omega$ is always well defined, but its value on an element of M_{sa} need not a priori be represented by a bounded element of M. The following recursive formalism is introduced to retain precisely the additional von Neumann algebraic data needed to guarantee this bounded realizability. The construction is related to projectional skeletons and projectional resolutions of the identity in nonseparable Banach space theory [Kub09]: both organize a large space by compatible bounded projections onto smaller pieces. The present construction is more rigid and more specialized. Every projection occurring in a certificate is required to be computable by finitely many trace-preserving conditional expectations, and its value on M_{sa} must remain in M_{sa}. The word "finite" below refers only to the depth of the certificate tree; the individual layers may have arbitrary cardinality.

Definition 4.3 (Finite-layer certificates and certified backgrounds)**.** A *finite-layer certificate* $\mathfrak{c}$ is defined recursively, together with its underlying orthonormal family

$$\Omega(\mathfrak{c}) \subset \mathcal{S}_0(M)$$

and, for every $x = x^* \in M$, a bounded self-adjoint element $P_{\mathfrak{c}}(x) \in M_{\mathrm{sa}}$.

The empty certificate. There is an empty certificate $\varnothing$ with

$$\Omega(\varnothing) = \varnothing, \qquad P_{\varnothing}(x) = 0.$$

Adding one layer. Suppose that $\mathfrak{c}^-$ is a finite-layer certificate with

$$\Omega^- = \Omega(\mathfrak{c}^-).$$

Suppose also that $\mathfrak{c}_\Theta$ is a finite-layer certificate with

$$\Theta = \Omega(\mathfrak{c}_\Theta) \subset \Omega^-.$$

Let $U \subset \mathcal{S}_0(M)$ be an orthonormal family which is orthogonal to Ω^-. Assume that there are von Neumann subalgebras

$$C \subset R \subset M$$

such that

$$L^2(R)_{\mathrm{sa}} = L^2(C)_{\mathrm{sa}} \oplus \overline{\operatorname{span}_{\mathbb{R}}}(\Theta \cup U).$$

Then the finite data

$$\mathfrak{c} = \left(\mathfrak{c}^-, \mathfrak{c}_\Theta, C, R, U\right)$$

is a finite-layer certificate with underlying family

$$\Omega(\mathfrak{c}) = \Omega^- \dot{\cup} U.$$

We say that U is the *new layer* of $\mathfrak{c}$, and that it is a layer over the certified subbackground $(\Theta, \mathfrak{c}_\Theta)$.

For $x = x^* \in M$, define

$$P_U^{\mathfrak{c}}(x) = \big(E_R(x) - E_C(E_R(x))\big) - P_{\mathfrak{c}_\Theta}(x),$$

and

$$P_{\mathfrak{c}}(x) = P_{\mathfrak{c}^-}(x) + P_U^{\mathfrak{c}}(x).$$

A *finite-layer certified background* is a pair $(\Omega, \mathfrak{c})$, where $\mathfrak{c}$ is a finite-layer certificate and $\Omega = \Omega(\mathfrak{c})$. When the certificate is fixed, we suppress it from the notation and write simply $P_\Omega(x)$ for $P_{\mathfrak{c}}(x)$.

A subfamily $\Theta \subset \Omega$ is called a *certified subbackground* when it is equipped with some finite-layer certificate $\mathfrak{c}_\Theta$ satisfying $\Omega(\mathfrak{c}_\Theta) = \Theta$. This certificate need not be a literal subcertificate of the fixed certificate for Ω; it is part of the finite data being used. The adjective *finite-layer* refers only to the finite depth of the certificate. The individual layers may have arbitrary cardinality.

To make the inductions on certificates formally unambiguous, we attach to each certificate a finite complexity rank. This rank is only a bookkeeping device for the recursive syntax introduced above and does not impose any cardinality restriction on the layers.

Remark 4.4. We define the complexity of a finite-layer certificate recursively by $\ell(\varnothing) = 0$, and, for $\mathfrak{c} = (\mathfrak{c}^-, \mathfrak{c}_\Theta, C, R, U)$, by

$$\ell(\mathfrak{c}) = 1 + \ell(\mathfrak{c}^-) + \ell(\mathfrak{c}_\Theta).$$

All inductions on finite-layer certificates below are taken with respect to this finite complexity.

The next lemma is the verification statement for the certificate formalism. Its proof is a finite induction based on Lemma 4.2, but its conclusion is essential: the recursively defined formula is not merely a formal expression in conditional expectations; it is exactly the Hilbert-space projection onto the underlying certified family and is represented by an element of M_{sa}.

Lemma 4.5. *Let Ω be a finite-layer certified background. Then for every $x = x^* \in M$, the Hilbert-space projection $P_\Omega x$ onto $\overline{\mathrm{span}_{\mathbb{R}}}\Omega$ belongs to M_{sa}. More precisely, $P_\Omega x$ is a finite sum of expressions obtained from x by trace-preserving conditional expectations appearing in the certificate.*

Proof. We prove the assertion by induction on the complexity of the finite certificate. The empty certificate is trivial.

Let $\mathfrak{c} = \big(\mathfrak{c}^-, \mathfrak{c}_\Theta, C, R, U\big)$ be a certificate obtained by adding one layer. Put

$$\Omega^- = \Omega(\mathfrak{c}^-), \qquad \Theta = \Omega(\mathfrak{c}_\Theta), \qquad \Omega = \Omega^- \dot{\cup} U.$$

By the induction hypothesis, $P_{\mathfrak{c}^-}(x)$ is the Hilbert-space projection of x onto $\overline{\mathrm{span}_{\mathbb{R}}}\Omega^-$, and $P_{\mathfrak{c}_\Theta}(x)$ is the Hilbert-space projection of x onto $\overline{\mathrm{span}_{\mathbb{R}}}\Theta$. Both belong to M_{sa}.

The completed-block decomposition

$$L^2(R)_{\mathrm{sa}} = L^2(C)_{\mathrm{sa}} \oplus \overline{\mathrm{span}_{\mathbb{R}}}(\Theta \cup U)$$

and the projection formula (4.1) show that

$$E_R(x) - E_C(E_R(x))$$

is the Hilbert-space projection of x onto $\overline{\mathrm{span}_{\mathbb{R}}}(\Theta \cup U)$. Since $U \perp \Omega^-$ and $\Theta \subset \Omega^-$, this projection splits orthogonally as the sum of the projection onto $\overline{\mathrm{span}_{\mathbb{R}}}\Theta$ and the projection onto $\overline{\mathrm{span}_{\mathbb{R}}}U$. Hence

$$P_U^{\mathfrak{c}}(x) = \big(E_R(x) - E_C(E_R(x))\big) - P_{\mathfrak{c}_\Theta}(x)$$

is exactly the Hilbert-space projection of x onto $\overline{\operatorname{span}_{\mathbb{R}}} U$. Therefore

$$P_{\mathfrak{c}}(x) = P_{\mathfrak{c}^-}(x) + P_U^{\mathfrak{c}}(x)$$

is the Hilbert-space projection of x onto $\overline{\operatorname{span}_{\mathbb{R}}}\Omega$. The displayed formula is a finite combination of trace-preserving conditional expectations and previously constructed projection formulas, so it belongs to M_{sa}. □

Consequently, if two finite-layer certificates have the same underlying orthonormal family Θ, then their formulas give the same element of M_{sa} for every $x = x^* \in M$, since both compute the Hilbert-space projection of x onto $\overline{\operatorname{span}_{\mathbb{R}}}\Theta$.

5. The strong closure lemma for certified backgrounds

We illustrate the purpose of the closure construction. The closure operation below is used only to ensure the following point: whenever a small subfamily of a certified background is selected, it can be enlarged to a still small subfamily on which all relevant projection supports are closed. Consequently the Hilbert-space projections onto these enlarged subfamilies are represented by bounded elements of the ambient von Neumann algebra M. No additional model-theoretic property of M is being assumed.

We begin with a standard Hilbert-space observation. Even when an orthonormal family is uncountable, every individual vector has at most countably many nonzero coordinates with respect to that family. This elementary fact is the reason that the support data used in the Skolem-hull construction remain countable.

Lemma 5.1. *Let T be an orthonormal family in a Hilbert space H. For every $\xi \in H$,*

$$\operatorname{supp}_T(\xi) = \{t \in T : \langle \xi, t\rangle \neq 0\}$$

is countable.

Proof. For $n \geq 1$, the set

$$F_n = \{t \in T : |\langle \xi, t\rangle| \geq 1/n\}$$

is finite by Bessel's inequality. Since

$$\operatorname{supp}_T(\xi) = \bigcup_{n=1}^{\infty} F_n,$$

the support is countable. □

Here and below, for a regular cardinal χ, $H(\chi)$ denotes the set of all sets whose transitive closure has cardinality $< \chi$. We use only the standard elementary consequences of closing $H(\chi)$ under countably many finitary Skolem functions.

We shall use the classical elementary-submodel method for $H(\chi)$. The following facts are standard consequences of Skolemization in a countable first-order language; see [CK90, Jech03]. Closely related elementary-submodel techniques are widely used for separable reduction and for the construction of bounded projections in functional analysis; see, for example, [Kub09, CK14]. We record the precise form needed here in order to fix the cardinal and chain-continuity conventions.

Lemma 5.2 (Skolem-hull closure fact)**.** *Let χ be regular and let $\mathfrak{A}$ be a countable first-order expansion of $(H(\chi), \in)$ by finitary function symbols, including Skolem functions, constants for ω, and constants for all natural numbers. For $S \subset H(\chi)$, let* $\operatorname{Hull}_{\mathfrak{A}}(S)$ *denote the Skolem hull generated by S together with the named constants. Then:*

(i) $|\operatorname{Hull}_{\mathfrak{A}}(S)| \leq |S| + \aleph_0$*;*

(ii) $S_1 \subset S_2$ *implies* $\operatorname{Hull}_{\mathfrak{A}}(S_1) \subset \operatorname{Hull}_{\mathfrak{A}}(S_2)$*;*

(iii) for every increasing chain $(S_i)_{i\in I}$*,*

$$\mathrm{Hull}_{\mathfrak{A}}\Big(\bigcup_i S_i\Big) = \bigcup_i \mathrm{Hull}_{\mathfrak{A}}(S_i);$$

(iv) if $c \in \mathrm{Hull}_{\mathfrak{A}}(S)$ *is countable in the universe, then* $c \subset \mathrm{Hull}_{\mathfrak{A}}(S)$*.*

Proof. The first three assertions follow from the usual construction of the closure under countably many finitary Skolem functions; in the chain-continuity statement, every finite tuple from an increasing union is already contained in one member of the chain. For (iv), the hull is an elementary submodel of the Skolemized structure and contains every natural number. Since c is countable, $H(\chi)$ satisfies that there is a surjection $f : \omega \to c$. By elementarity and the Skolem functions, such an f belongs to the hull. For every $n \in \omega$, the value $f(n)$ is definable from f and n, and hence belongs to the hull. Thus $c \subset \mathrm{Hull}_{\mathfrak{A}}(S)$. This is the standard Skolem-hull argument; see, for example, [Jech03, CK90]. □

The elementary-submodel and projectional-skeleton methods in functional analysis provide general mechanisms for passing to small subspaces while preserving selected operations and projections; see [Kub09, CK14]. The next lemma is an operator-algebraic implementation of this principle for a single completed block.

The standard Skolem-hull ingredients alone do not give the conclusion needed here. The hull must also be closed under the relevant conditional expectations and under the support map of the completed-block projection. Countability of Hilbert-space supports then allows one to reconstruct smaller von Neumann algebras $A_S \subset A$ and $Q_S \subset Q$ for which the selected family S remains a complete orthogonal complement. Thus both the von Neumann algebraic structure and the bounded projection formula are preserved. To the best of our knowledge, this precise closed-subblock statement is new.

Lemma 5.3 (Closed subblocks of a completed block)**.** *Let* $\mathfrak{B} = (A, Q, T)$ *be a completed block. There is a closure operation* $\mathrm{cl}_{\mathfrak{B}}$ *on subsets of* T *with the following properties:*

(i) $S \subset \mathrm{cl}_{\mathfrak{B}}(S)$*;*
(ii) $\mathrm{cl}_{\mathfrak{B}}(\mathrm{cl}_{\mathfrak{B}}(S)) = \mathrm{cl}_{\mathfrak{B}}(S)$*;*
(iii) $|\mathrm{cl}_{\mathfrak{B}}(S)| \leq |S| + \aleph_0$*;*
(iv) $S_1 \subset S_2$ *implies* $\mathrm{cl}_{\mathfrak{B}}(S_1) \subset \mathrm{cl}_{\mathfrak{B}}(S_2)$*;*
(v) for every increasing chain $(S_i)_{i\in I}$*,*

$$\mathrm{cl}_{\mathfrak{B}}\Big(\bigcup_i S_i\Big) = \bigcup_i \mathrm{cl}_{\mathfrak{B}}(S_i);$$

(vi) if $S = \mathrm{cl}_{\mathfrak{B}}(S)$*, then there are von Neumann algebras* $A_S \subset A$ *and* $Q_S \subset Q$ *such that*

$$(A_S, Q_S, S)$$

is a completed block.

Proof. We spell out the Skolem-hull construction. Choose a regular cardinal χ such that

$$M, A, Q, T, \tau, E_A, E_Q, \omega,\ [T]^{\leq \aleph_0}$$

belong to $H(\chi)$, and such that the graphs of the algebraic operations on M also belong to $H(\chi)$. We take χ large enough for all these requirements throughout the argument below.

Work in a countable first-order expansion $\mathfrak{A}_{\mathfrak{B}}$ of $(H(\chi), \in)$. The objects M, A, Q, T, the algebraic operations on M, and the conditional expectations E_A, E_Q are named as single predicates, function symbols, or set-valued parameters. The algebraic operations include the constants 0_M and 1_M, addition, multiplication, adjoint, and scalar multiplication by each $\lambda \in \mathbb{Q} + i\mathbb{Q}$. We do not add constants for all individual elements of M, A, Q, or T.

We include a constant for ω and constants for every natural number. Hence every Skolem hull $\mathcal{E}$ considered below satisfies

$$\omega \in \mathcal{E} \qquad \text{and} \qquad \omega \subset \mathcal{E}.$$

Every partial map used below is made into a total function on $H(\chi)$ by assigning an arbitrary fixed value outside its intended domain.

As a function symbol in the expansion, include the total support map

$$\sigma_{\mathfrak{B}}(x) = \operatorname{supp}_T\big(E_Q(x) - E_A(E_Q(x))\big), \qquad x = x^* \in M,$$

again with an arbitrary fixed value outside M_{sa}. By Lemma 5.1, for $x = x^* \in M$ the value $\sigma_{\mathfrak{B}}(x)$ is a countable subset of T, and hence belongs to $H(\chi)$.

The language is countable, since only countably many algebraic operations, conditional expectations, support maps, and named structural parameters have been added. Add countably many finitary Skolem functions for this countable structure. For $S \subset T$, let $\mathrm{Hull}_{\mathfrak{B}}(S)$ be the Skolem hull generated by S, together with the fixed structural parameters just described, the constant ω, and the constants for all natural numbers. Define

$$\mathrm{cl}_{\mathfrak{B}}(S) = T \cap \mathrm{Hull}_{\mathfrak{B}}(S).$$

Thus the hull is generated from S, the natural numbers, and the fixed structural objects, but not from all individual elements of M, A, Q, or T.

By Lemma 5.2, the size, monotonicity, and chain-continuity assertions follow from the countability of the language and the finitarity of the Skolem functions. Idempotence follows because

$$\mathrm{Hull}_{\mathfrak{B}}(S) \subset \mathrm{Hull}_{\mathfrak{B}}(\mathrm{cl}_{\mathfrak{B}}(S)) \subset \mathrm{Hull}_{\mathfrak{B}}(\mathrm{Hull}_{\mathfrak{B}}(S)) = \mathrm{Hull}_{\mathfrak{B}}(S).$$

Assume now that $S = \mathrm{cl}_{\mathfrak{B}}(S)$, and set $\mathcal{E} = \mathrm{Hull}_{\mathfrak{B}}(S)$. Then $S = T \cap \mathcal{E}$. We shall use the standard elementary-hull fact: if $c \in \mathcal{E}$ is countable and $\omega \subset \mathcal{E}$, then $c \subset \mathcal{E}$. Indeed, elementarity gives a surjection $f : \omega \to c$ with $f \in \mathcal{E}$, and then $f(n) \in \mathcal{E}$ for every $n \in \omega$.

Define

$$A_S = W^*(A \cap \mathcal{E}), \qquad Q_S = W^*(A_S, S).$$

We show that (A_S, Q_S, S) is a completed block. Let $w = w(a_1, \ldots, a_m, s_1, \ldots, s_n)$ be a bounded self-adjoint $*$-word with coefficients in $\mathbb{Q} + i\mathbb{Q}$, where $a_i \in A \cap \mathcal{E}$ and $s_j \in S$. Since $S \subset T \subset Q$ and $A \subset Q$, we have $w \in Q$. Moreover, because the algebraic operations are included in the Skolemized structure, $w \in \mathcal{E}$. In the original block,

$$w = E_A(w) + P_T^{\mathfrak{B}}(w).$$

Since E_A is part of the structure, $E_A(w) \in A \cap \mathcal{E}$. Moreover

$$\operatorname{supp}_T(P_T^{\mathfrak{B}}(w)) = \sigma_{\mathfrak{B}}(w) \in \mathcal{E}.$$

This set is countable. Since $\mathcal{E}$ contains ω and all natural numbers, Lemma 5.2*(iv)* gives

$$\sigma_{\mathfrak{B}}(w) \subset \mathcal{E}.$$

Hence

$$\sigma_{\mathfrak{B}}(w) \subset T \cap \mathcal{E} = S,$$

and therefore

$$P_T^{\mathfrak{B}}(w) \in \overline{\operatorname{span}_{\mathbb{R}}} S.$$

Let $\mathcal{A}_S$ be the unital rational-coefficient algebraic $*$-algebra generated by $A \cap \mathcal{E}$ and S, and let $B_S = \overline{\mathcal{A}_S}^{\|\cdot\|}$ be its norm closure. Then

$$Q_S = W^*(A_S, S) = W^*(A \cap \mathcal{E}, S) = B_S''.$$

Kaplansky's density theorem [Tak02, p. 82, Theorem II.4.8] applies to the C^*-algebra B_S: the self-adjoint part of the unit ball of B_S is strong-$*$ dense in the self-adjoint part of the unit ball of

Q_S. Since $\mathcal{A}_S$ is norm dense in B_S, and since bounded strong convergence implies L^2-convergence in a finite von Neumann algebra, the bounded self-adjoint rational-coefficient $*$-words in $A \cap \mathcal{E}$ and S are L^2-dense in $L^2(Q_S)_{\mathrm{sa}}$. Hence

$$L^2(Q_S)_{\mathrm{sa}} \subset L^2(A_S)_{\mathrm{sa}} + \overline{\mathrm{span}_{\mathbb{R}}} S.$$

The reverse inclusion is clear from the definition of Q_S, and the sum is orthogonal because $S \perp L^2(A)_{\mathrm{sa}}$ and $A_S \subset A$. Thus

$$L^2(Q_S)_{\mathrm{sa}} = L^2(A_S)_{\mathrm{sa}} \oplus \overline{\mathrm{span}_{\mathbb{R}}} S.$$

$\square$

Lemma 5.3 treats a single completed block. The next proposition propagates that closure construction through an arbitrary finite certificate tree. This requires simultaneous control of the old certified background, the certified support over which a new layer is attached, and the completed block witnessing that layer.

The construction is conceptually analogous to the coherence requirements in projectional-skeleton methods [Kub09], but here the closed pieces must retain finite-layer certificates and their projections must remain expressible by conditional expectations in M. A further essential point is compatibility under adjoining a new layer: taking a closed subfamily of the enlarged background must recover a closed subfamily of the old background. This compatibility will be used at every successor stage of the all-cardinal extension argument. The proposition is one of the principal new technical ingredients of the paper.

Proposition 5.4 (Strong closure for finite-layer backgrounds)**.** *Let Ω be a finite-layer certified background with a fixed finite certificate. There is a closure operation $\mathrm{cl}_\Omega : \mathcal{P}(\Omega) \to \mathcal{P}(\Omega)$ such that:*

(i) $S \subset \mathrm{cl}_\Omega(S)$*;*
(ii) $\mathrm{cl}_\Omega(\mathrm{cl}_\Omega(S)) = \mathrm{cl}_\Omega(S)$*;*
(iii) $|\mathrm{cl}_\Omega(S)| \le |S| + \aleph_0$*;*
(iv) $S_1 \subset S_2$ *implies* $\mathrm{cl}_\Omega(S_1) \subset \mathrm{cl}_\Omega(S_2)$*;*
(v) for every increasing chain $(S_i)_{i\in I}$*,* $\mathrm{cl}_\Omega\left(\bigcup_i S_i\right) = \bigcup_i \mathrm{cl}_\Omega(S_i)$;
(vi) if $S = \mathrm{cl}_\Omega(S)$*, then* S *is itself a finite-layer certified background. Consequently* $P_S(x) \in M_{\mathrm{sa}}$ *for every* $x = x^* \in M$*.*

Moreover the closure operations may be chosen compatibly with adding one more layer: if U is a new layer over a cl_Ω-closed certified subbackground $\Theta \subset \Omega$ and $\Omega^+ = \Omega \dot\cup U$, then cl_{Ω^+} can be chosen so that every cl_{Ω^+}-closed $\Xi \subset \Omega^+$ satisfies

$$\Xi \cap \Omega = \mathrm{cl}_\Omega(\Xi \cap \Omega).$$

Proof. We prove the closure and compatibility assertions simultaneously by induction on the complexity of the fixed finite certificate.

Fix once and for all a regular cardinal χ large enough that the entire finite certificate tree for Ω, all completed-block data occurring in that tree, all relevant conditional expectations and support maps, and the graphs of the algebraic operations on M belong to $H(\chi)$. All Skolem hulls used in this proof are taken inside this same structure $H(\chi)$.

At each node of the certificate tree, we use a countable expansion of $H(\chi)$. When passing to a later node, we enlarge the language by the structural data and the Skolem functions used at the earlier nodes. Every partial operation is extended to a total function on $H(\chi)$ by assigning a fixed value outside its natural domain. Since the certificate tree has finite complexity, only finitely many previously constructed languages occur, and their union is still countable. Thus every previously used Skolem function is a total function on the same underlying set $H(\chi)$, and the closure operations used below are formally compatible.

The empty certificate is trivial. Suppose that

$$\Omega = \Omega^{-} \dot{\cup} U,$$

where Ω^{-} has already been equipped with a closure operation having the required properties, and where U is a layer over a certified subbackground $\Theta \subset \Omega^{-}$. Thus there are von Neumann algebras $C \subset R \subset M$ such that

$$L^2(R)_{\mathrm{sa}} = L^2(C)_{\mathrm{sa}} \oplus \overline{\operatorname{span}_{\mathbb{R}}}(\Theta \cup U).$$

Use a countable Skolemized expansion of the fixed structure $H(\chi)$ containing:

- all data and Skolem functions used for the closure operations already constructed on Ω^{-} and on the certified subbackgrounds occurring in its finite certificate, in particular on Θ;
- the completed-block data $(C, R, \Theta \cup U)$ and the corresponding support map from Lemma 5.3;
- the algebraic operations on M, the relevant conditional expectations, and constants for ω and all natural numbers.

Since the fixed certificate has finite complexity, only finitely many previously constructed closure languages occur here; their union is still a countable language.

For $S \subset \Omega$, let $\mathrm{Hull}_{\Omega}(S)$ be the hull of $S \cup \{\omega\}$ together with the structural parameters in this structure, and define

$$\mathrm{cl}_{\Omega}(S) = \Omega \cap \mathrm{Hull}_{\Omega}(S).$$

Lemma 5.2 gives the size, monotonicity, and chain-continuity assertions, and the same hull calculation as in Lemma 5.3 gives idempotence.

Assume $S = \mathrm{cl}_{\Omega}(S)$ and put $\mathcal{E} = \mathrm{Hull}_{\Omega}(S)$. Then $S = \Omega \cap \mathcal{E}$. Let

$$S^{-} = S \cap \Omega^{-}, \qquad U_S = S \cap U, \qquad \Theta_S = \Theta \cap \mathcal{E}.$$

Because the Skolem functions defining $\mathrm{cl}_{\Omega^{-}}$ are included in the present language, S^{-} is $\mathrm{cl}_{\Omega^{-}}$-closed. Indeed,

$$\mathrm{cl}_{\Omega^{-}}(S^{-}) \subset \Omega^{-} \cap \mathcal{E} = S^{-},$$

and the reverse inclusion follows from the definition of a closure operation. Hence S^{-} is certified by the induction hypothesis. Let cl_{Θ} be a strong closure operation for the certified background Θ, supplied by the induction hypothesis for the finite certificate of Θ. The Skolem functions defining cl_{Θ} have been included in the present language. Therefore

$$\mathrm{cl}_{\Theta}(\Theta_S) \subset \Theta \cap \mathcal{E} = \Theta_S,$$

and the reverse inclusion follows from extensivity of the closure operation. Thus Θ_S is cl_{Θ}-closed, and hence is a certified subbackground. Since $\Theta \subset \Omega^{-}$, we also have $\Theta_S \subset S^{-}$. Because the Skolem functions for the completed block $\mathfrak{B} = (C, R, \Theta \cup U)$ are included in the present language,

$$T_S = (\Theta \cup U) \cap \mathcal{E} = \Theta_S \cup U_S$$

is closed for the block closure from Lemma 5.3. Indeed,

$$\mathrm{cl}_{\mathfrak{B}}(T_S) \subset (\Theta \cup U) \cap \mathcal{E} = T_S,$$

and the reverse inclusion is automatic. Applying that lemma to this closed set gives von Neumann algebras $C_S \subset C$ and $R_S \subset R$ such that

$$L^2(R_S)_{\mathrm{sa}} = L^2(C_S)_{\mathrm{sa}} \oplus \overline{\operatorname{span}_{\mathbb{R}}}(\Theta_S \cup U_S).$$

Thus U_S is a layer over the certified subbackground $\Theta_S \subset S^{-}$. It follows that

$$S = S^{-} \dot{\cup} U_S$$

is finite-layer certified.

For the final compatibility assertion, construct cl_{Ω^+} by using a Skolem language which contains all Skolem functions used for cl_Ω and the completed-block data defining the new layer U over Θ. Let $\Xi = \mathrm{cl}_{\Omega^+}(\Xi)$ and $\mathcal{E} = \mathrm{Hull}_{\Omega^+}(\Xi)$. Then

$$\Xi = \Omega^+ \cap \mathcal{E}, \qquad \Xi \cap \Omega = \Omega \cap \mathcal{E}.$$

Since $\mathcal{E}$ is closed under the finitary Skolem functions defining cl_Ω,

$$\mathrm{cl}_\Omega(\Xi \cap \Omega) \subset \Omega \cap \mathcal{E} = \Xi \cap \Omega.$$

The reverse inclusion follows from extensivity, and therefore

$$\Xi \cap \Omega = \mathrm{cl}_\Omega(\Xi \cap \Omega).$$

We also need the old support of the new layer to be certified. Since $\Theta = \mathrm{cl}_\Omega(\Theta)$ and $\mathcal{E}$ is closed under the Skolem functions defining cl_Ω, we have

$$\mathrm{cl}_\Omega(\Theta \cap \mathcal{E}) \subset \mathrm{cl}_\Omega(\Theta) \cap \mathcal{E} = \Theta \cap \mathcal{E}.$$

Again the reverse inclusion follows from extensivity, so

$$\Theta \cap \mathcal{E} = \mathrm{cl}_\Omega(\Theta \cap \mathcal{E}).$$

Thus $\Theta \cap \mathcal{E}$ is a certified subbackground by the closure assertions already proved for Ω.

Finally, apply Lemma 5.3 to the completed block $(C, R, \Theta \cup U)$. Since the corresponding block-closure Skolem functions are included in the language,

$$(\Theta \cup U) \cap \mathcal{E} = (\Theta \cap \mathcal{E}) \cup (\Xi \cap U)$$

is closed for that block closure. Hence there are von Neumann algebras $C_\Xi \subset C$ and $R_\Xi \subset R$ such that

$$L^2(R_\Xi)_{\mathrm{sa}} = L^2(C_\Xi)_{\mathrm{sa}} \oplus \overline{\mathrm{span}_{\mathbb{R}}}\big((\Theta \cap \mathcal{E}) \cup (\Xi \cap U)\big).$$

Thus $\Xi \cap U$ is a certified layer over the certified subbackground $\Theta \cap \mathcal{E}$. Since

$$\Theta \cap \mathcal{E} \subset \Xi \cap \Omega$$

and $\Xi \cap \Omega$ is cl_Ω-closed, the closure assertions already proved for Ω show that $\Xi \cap \Omega$ is finite-layer certified. Therefore

$$\Xi = (\Xi \cap \Omega) \dot\cup (\Xi \cap U)$$

is finite-layer certified by adjoining the layer $\Xi \cap U$ over the certified subbackground $\Theta \cap \mathcal{E} \subset \Xi \cap \Omega$. $\square$

For later applications, we isolate the following formal consequences of Proposition 5.4 and the recursive definition of a certificate. The corollary introduces no additional construction; its purpose is to provide the precise restriction and adjoining operations used in the transfinite induction.

Corollary 5.5. *Let $(\Omega, \mathfrak{c}_\Omega)$ be a finite-layer certified background, and let cl_Ω be a closure operation supplied by Proposition 5.4.*

(i) If $S \subset \Omega$ satisfies $S = \mathrm{cl}_\Omega(S)$, then S admits a finite-layer certificate $\mathfrak{c}_S$. With this certificate, $P_{\mathfrak{c}_S}(x)$ is the Hilbert-space projection of x onto $\overline{\mathrm{span}_{\mathbb{R}}} S$ for every $x = x^* \in M$.

(ii) Suppose $S = \mathrm{cl}_\Omega(S)$ and $U \subset \mathcal{S}_0(M)$ is an orthonormal family orthogonal to Ω. Suppose further that there are von Neumann algebras $C \subset R \subset M$ such that

$$L^2(R)_{\mathrm{sa}} = L^2(C)_{\mathrm{sa}} \oplus \overline{\mathrm{span}_{\mathbb{R}}}(S \cup U).$$

Then $\Omega \dot\cup U$ is a finite-layer certified background. A certificate is obtained by taking the old certificate $\mathfrak{c}_\Omega$, the certificate $\mathfrak{c}_S$ from part (i)*, and adjoining the single layer U over S.*

(iii) The strong closure operation on $\Omega \dot\cup U$ may be chosen compatibly with cl_Ω in the sense of Proposition 5.4.

Proof. Part (i) is exactly Proposition 5.4(vi), together with Lemma 4.5. Part (ii) follows directly from the recursive definition of finite-layer certificates in Definition 4.3. Part (iii) is the compatibility assertion in Proposition 5.4. □

6. The countable certified extension

We first prove the extension theorem in the countable case. Its analytic descent mechanism is modeled on the greedy norm-reduction argument and diagonalization used in [HTZ26, Claim 3 and the proof of Theorem 1]. In the present construction, however, a task must not merely be revisited infinitely often: its visiting times n_k must satisfy $\sum_k n_k^{-1} = \infty$, because the available distance decrease is of harmonic order. The following elementary strengthening of the usual dovetailing argument supplies exactly this schedule.

Lemma 6.1. *Suppose that a recursion is indexed by $n \geq 1$, and that at each stage at most countably many new tasks may appear. Then the tasks can be labelled and scheduled so that every task which eventually appears is processed infinitely often at times n_k satisfying*

$$\sum_k \frac{1}{n_k} = \infty.$$

Proof. For $j \in \{0, 1, 2, \ldots\}$, put

$$A_j = \{n \geq 1 : \nu_2(n) = j\},$$

where $\nu_2(n)$ denotes the largest ν such that $2^\nu \mid n$. Then the sets A_j partition the positive integers, and

$$\sum_{n \in A_j} \frac{1}{n} = \infty \qquad (j = 0, 1, 2, \ldots).$$

Next partition the label set $\{0, 1, 2, \ldots\}$ into pairwise disjoint infinite subsets

$$\{0, 1, 2, \ldots\} = \bigsqcup_{m=0}^{\infty} L_m.$$

The labels in L_m will be reserved for tasks which first appear at stage m.

At stage m, the new batch of tasks is countable, so we assign those tasks injectively to unused labels in the infinite set L_m. At time $n \geq 1$, let $j = \nu_2(n)$. If a task has already been assigned label j, we process that task; otherwise we do nothing at time n.

If a task first appears at stage m and receives label $j \in L_m$, then it is processed at all sufficiently late times in A_j. Removing finitely many early terms from A_j does not change the divergence of the harmonic sum. Hence the visiting times n_k of this task satisfy

$$\sum_k \frac{1}{n_k} = \infty.$$

□

Inspired by the argument in [HTZ26, Claim 3], we obtain the following lemma.

Lemma 6.2. *Let $b = b^* \in M$, and let $V_1 \subset V_2 \subset \cdots \subset L^2(M, \tau)_{\mathrm{sa}}$ be finite-dimensional real subspaces. Assume that V_n is spanned by at most n orthonormal symmetries. Let $\mathcal{N} = \{n_1 < n_2 < \cdots\} \subset \mathbb{N}$ satisfy $\sum_{k=1}^{\infty} 1/n_k = \infty$. Suppose that, for every $n \in \mathcal{N}$, whenever $r_n = b - P_{V_n} b \neq 0$, there is a symmetry $u_n \in M$ such that*

$$u_n \perp V_n, \qquad V_{n+1} \supset V_n \oplus \mathbb{R} u_n, \qquad \langle r_n, u_n \rangle_2 = \frac{\|r_n\|_2^2}{\|r_n\|_\infty}.$$

Then

$$b \in \overline{\bigcup_{n\geq 1} V_n}^{\|\cdot\|_2}.$$

Proof. Set

$$D_n = \operatorname{dist}(b, V_n), \qquad d = \lim_{n\to\infty} D_n.$$

Suppose that $d > 0$. In particular, $b \neq 0$, so

$$K_b = \|b\|_\infty + \|b\|_2 > 0.$$

If V_n is spanned by orthonormal symmetries $v_1, \dots, v_{m_n}$, where $m_n \leq n$, then Bessel's inequality gives

$$\begin{aligned}\|b - P_{V_n} b\|_\infty &\leq \|b\|_\infty + \sum_{j=1}^{m_n} |\langle b, v_j\rangle_2| \\ &\leq \|b\|_\infty + \sqrt{m_n}\,\|b\|_2 \\ &\leq K_b \sqrt{n}.\end{aligned}$$

At a visiting time n_k, the vector u_{n_k} is orthogonal to V_{n_k}, and therefore

$$\begin{aligned}D_{n_k+1}^2 &\leq \operatorname{dist}\big(b, V_{n_k} + \mathbb{R}u_{n_k}\big)^2 \\ &= D_{n_k}^2 - |\langle r_{n_k}, u_{n_k}\rangle_2|^2 \\ &= D_{n_k}^2 - \frac{\|r_{n_k}\|_2^4}{\|r_{n_k}\|_\infty^2} \\ &\leq D_{n_k}^2 - \frac{d^4}{K_b^2 n_k}.\end{aligned}$$

Since D_n is decreasing,

$$D_{n_{k+1}}^2 \leq D_{n_k+1}^2.$$

Consequently, for every $\ell \geq 1$,

$$\sum_{k=1}^{\ell} \frac{d^4}{K_b^2 n_k} \leq D_{n_1}^2.$$

Letting $\ell \to \infty$ contradicts

$$\sum_{k=1}^{\infty} \frac{1}{n_k} = \infty.$$

Thus $d = 0$. □

We now combine the relative norming lemma, the sparse greedy descent argument, and the certificate machinery to prove Lemma 6.3. The analytic part of the construction has been isolated in Lemma 6.2. At each prescribed visiting time, the relative norming lemma provides a new symmetry whose correlation with the current residual gives the required decrease of the squared distance. Lemma 6.1 ensures that every eventual task is revisited along a sequence of stages (n_k) satisfying

$$\sum_k \frac{1}{n_k} = \infty,$$

so that Lemma 6.2 forces the corresponding residual distance to vanish.

The remaining issue is to make this descent compatible with the operator-algebraic recursion. During the construction the parameter set grows, and one must treat every self-adjoint

$*$-polynomial word that eventually appears. Moreover, the Hilbert-space projection onto the prescribed background must be represented by a bounded element of M. This is precisely where the certificate hypothesis is used. For an eventual self-adjoint word y, the certified projection $P_\Omega(y)$ belongs to M_{sa}, and hence

$$b(y) = y - E_N(y) - P_\Omega(y)$$

is a bounded self-adjoint element of M. After removing its component in the span of the previously chosen symmetries, the resulting residual satisfies the hypotheses of Lemma 3.4.

For comparison, De and Mukherjee [DM23, Theorem 3.4] proved a basis-extension result for uniformly bounded self-adjoint operators in separable GNS spaces. The countable extension lemma below has a different purpose: it preserves orthogonality simultaneously to a prescribed von Neumann subalgebra and to a possibly uncountable certified background, absorbs specified algebraic data, and produces a completed block whose associated projections remain represented inside M.

Lemma 6.3 (Countable certified extension)**.** *Let $\kappa = \operatorname{dens} L^2(M) > \aleph_0$. Let $N \subset M$ be a von Neumann subalgebra with $\operatorname{dens} L^2(N) < \kappa$. Let $\Omega \subset \mathcal{S}_0(M)$ be a finite-layer certified background with $|\Omega| < \kappa$, $E_N(\Omega) = 0$, and a fixed strong closure operation cl_Ω as in Proposition 5.4. Let $S \subset \Omega$, $X \subset M_{\mathrm{sa}}$, and $Y \subset N_{\mathrm{sa}}$ be countable. Then there are von Neumann algebras $C \subset N$ and $C \subset R \subset M$, a cl_Ω-closed set $\Omega_0 \subset \Omega$, and an orthonormal family $U \subset \mathcal{S}_0(M)$ such that*

$$S \subset \Omega_0, \qquad X \subset R, \qquad Y \subset C,$$
$$|\Omega_0|,\ |U|,\ \operatorname{dens} L^2(R) \le \aleph_0,$$
$$E_N(U) = 0, \qquad U \perp \Omega,$$

and

$$L^2(R)_{\mathrm{sa}} = L^2(C)_{\mathrm{sa}} \oplus \overline{\operatorname{span}_{\mathbb{R}}}(\Omega_0 \cup U).$$

Consequently U is a layer over Ω_0.

Proof. We run a countable recursive construction with fair bookkeeping. Maintain countable sets of parameters $\mathcal{C} \subset N_{\mathrm{sa}}$, $\Omega_* \subset \Omega$, and a finite or countable orthonormal family of already chosen symmetries U_*. Initially

$$\mathcal{C} = Y, \qquad \Omega_* = \mathrm{cl}_\Omega(S), \qquad U_* = \varnothing.$$

We implement the task list in countable batches, using Lemma 6.1.

By a formal $*$-polynomial word we mean a noncommutative $*$-polynomial with coefficients in $\mathbb{Q} + i\mathbb{Q}$ in finitely many parameters which have appeared by the relevant stage. For such a word w, we use the self-adjoint parts

$$\Re w = \frac{w + w^*}{2}, \qquad \Im w = \frac{w - w^*}{2i}.$$

An *eventual self-adjoint word* is an evaluated element of M_{sa} which is equal to $\Re w$ or $\Im w$ for some such formal word w, after all parameters occurring in w have appeared at some finite stage of the construction.

At stage 0, declare all elements of the initial parameter set

$$\mathcal{C} \cup X \cup \Omega_* \cup U_*$$

to have appeared. Enumerate all formal $*$-polynomial words with coefficients in $\mathbb{Q}+i\mathbb{Q}$ using these appeared parameters, take their real and imaginary self-adjoint parts, and regard the resulting self-adjoint words as the first countable batch of tasks.

At the end of each later stage, after $\mathcal{C}$, Ω_*, or U_* has possibly been enlarged, declare all newly added parameters to have appeared. Then enumerate all formal $*$-polynomial words with coefficients in $\mathbb{Q} + i\mathbb{Q}$ using the appeared parameters, take their real and imaginary self-adjoint parts, and add precisely those self-adjoint words which have not previously been labelled as the next countable batch of tasks.

By Lemma 6.1, the labels can be assigned so that every eventual self-adjoint word is processed infinitely many times at visiting times n_k satisfying

$$\sum_k \frac{1}{n_k} = \infty.$$

At a stage whose scheduled label has not yet been assigned, we perform no processing and continue to the next stage. Otherwise the scheduled label corresponds to a bounded self-adjoint word over the appeared parameter set, and we process that word as follows.

Thus suppose that the current scheduled task is represented by a bounded self-adjoint word $y \in M_{\mathrm{sa}}$. Define

$$b(y) = y - E_N(y) - P_\Omega(y). \tag{6.1}$$

The residual (6.1) is represented by a bounded self-adjoint element of M: indeed, $E_N(y) \in N_{\mathrm{sa}} \subset M_{\mathrm{sa}}$, and $P_\Omega(y) \in M_{\mathrm{sa}}$ by Lemma 4.5.

Moreover,

$$E_N(P_\Omega(y)) = 0,$$

because $P_\Omega(y)$ is an L^2-limit of finite real linear combinations of elements of Ω, while E_N is L^2-continuous and vanishes on Ω. Hence

$$E_N(b(y)) = 0.$$

Also

$$b(y) \perp \overline{\operatorname{span}_{\mathbb{R}}}\Omega.$$

Indeed, $y - P_\Omega(y)$ is orthogonal to $\overline{\operatorname{span}_{\mathbb{R}}}\Omega$ by definition of P_Ω, and $E_N(y) \perp \Omega$ because $E_N(y) \in N_{\mathrm{sa}}$ and $E_N(\Omega) = 0$.

For $\xi \in \overline{\operatorname{span}_{\mathbb{R}}}\Omega$, write

$$\operatorname{supp}_\Omega(\xi) = \{\omega \in \Omega : \langle \xi, \omega \rangle_2 \neq 0\}.$$

Add $E_N(y)$ to $\mathcal{C}$, add the countable set $\operatorname{supp}_\Omega(P_\Omega(y))$ to Ω_*, and then replace Ω_* by $\operatorname{cl}_\Omega(\Omega_*)$. This keeps Ω_* countable by Proposition 5.4.

At this finite stage, only finitely many symmetries have been chosen; write them as $u_1, \dots, u_m$. Since $b(y)$ and the u_i's are self-adjoint, the real Hilbert-space projection is

$$P_{\operatorname{span}_{\mathbb{R}}\{u_1,\dots,u_m\}} b(y) = \sum_{i=1}^m \langle b(y), u_i \rangle_2 u_i,$$

with real coefficients, and hence belongs to M_{sa}. Put

$$r = b(y) - P_{\operatorname{span}_{\mathbb{R}}\{u_1,\dots,u_m\}} b(y).$$

Thus $r \in M_{\mathrm{sa}}$. We record explicitly that r satisfies the hypotheses needed for Lemma 3.4. Since $E_N(b(y)) = 0$ and $E_N(u_i) = 0$ for $1 \le i \le m$, we have

$$E_N(r) = 0.$$

Moreover $b(y) \perp \Omega$ and $u_i \perp \Omega$ for all i, hence

$$\tau(\omega r) = 0 \qquad (\omega \in \Omega).$$

By construction r is the residual after orthogonal projection onto $\operatorname{span}_{\mathbb{R}}\{u_1, \dots, u_m\}$, so

$$\tau(u_i r) = 0 \qquad (1 \le i \le m).$$

Therefore r is orthogonal to

$$\Gamma = \Omega \cup \{u_1, \dots, u_m\}.$$

If $r = 0$, choose no new symmetry at this stage. If $r \neq 0$, apply Lemma 3.4 with this constraint set Γ. This set has cardinal $< \kappa$. We obtain $u_{m+1} \in \mathcal{S}_0(M)$ satisfying

$$E_N(u_{m+1}) = 0, \qquad u_{m+1} \perp \Omega, \qquad u_{m+1} \perp u_i \quad (1 \leq i \leq m),$$

and

$$\langle r, u_{m+1} \rangle_2 = \frac{\|r\|_2^2}{\|r\|_\infty}.$$

Add u_{m+1} to U_*.

Let U be the family of all chosen symmetries, let Ω_0 be the union of the increasing sequence of values of Ω_*, let $C = W^*(\mathcal{C})$, and put

$$R = W^*(C, X, \Omega_0, U).$$

The sets Ω_0 and U are countable. Moreover, R is generated by countably many bounded elements. Hence, by Lemma 3.1*(i)*,

$$\operatorname{dens} L^2(R) \leq \aleph_0.$$

The set Ω_0 is cl_Ω-closed by the continuity of the closure operation along the increasing sequence of values of Ω_*.

It remains to prove the completed-block decomposition. Fix an eventual self-adjoint word y. After all parameters occurring in y have appeared, the word receives a fixed label and is processed at visiting times

$$n_1 < n_2 < \cdots, \qquad \sum_k \frac{1}{n_k} = \infty,$$

by Lemma 6.1.

Let V_n be the real span of all symmetries chosen before stage n. At most one symmetry is chosen at each stage, so V_n is spanned by at most n orthonormal symmetries. Whenever y is processed and

$$r_n = b(y) - P_{V_n} b(y) \neq 0,$$

the construction adjoins a symmetry $u_n \perp V_n$ satisfying

$$\langle r_n, u_n \rangle_2 = \frac{\|r_n\|_2^2}{\|r_n\|_\infty}.$$

Lemma 6.2, applied to $b = b(y)$, therefore gives

$$b(y) \in \overline{\operatorname{span}_{\mathbb{R}}} U.$$

For such y, rearranging (6.1) gives

$$y = E_N(y) + P_\Omega(y) + b(y),$$

where $E_N(y) \in L^2(C)_{\mathrm{sa}}$, $P_\Omega(y) \in \overline{\operatorname{span}_{\mathbb{R}}} \Omega_0$ by the support bookkeeping, and $b(y) \in \overline{\operatorname{span}_{\mathbb{R}}} U$. The eventual words are L^2-dense in $L^2(R)_{\mathrm{sa}}$. Let $\mathcal{C}_\infty$ be the final value of $\mathcal{C}$, and put

$$\mathcal{P}_\infty = \mathcal{C}_\infty \cup X \cup \Omega_0 \cup U.$$

Then

$$C = W^*(\mathcal{C}_\infty), \qquad R = W^*(C, X, \Omega_0, U) = W^*(\mathcal{P}_\infty).$$

Every element of $\mathcal{P}_\infty$ appears at some finite stage of the construction. Hence every unital rational-coefficient $*$-word in $\mathcal{P}_\infty$, and its real and imaginary self-adjoint parts, is an eventual word. Let

$$\mathcal{A}_{\mathbb{Q}} = {}^*\text{-alg}_{\mathbb{Q}+i\mathbb{Q}}(\mathcal{P}_\infty)$$

denote the unital $*$-algebra over $\mathbb{Q}+i\mathbb{Q}$ generated by $\mathcal{P}_\infty$. The self-adjoint part of $\mathcal{A}_{\mathbb{Q}}$ is contained in the set of eventual self-adjoint words. Its norm closure is the C^*-algebra generated by $\mathcal{P}_\infty$, whose bicommutant is R. By Kaplansky density, $(\mathcal{A}_{\mathbb{Q}})_{\mathrm{sa}}$ is L^2-dense in $L^2(R)_{\mathrm{sa}}$. Therefore the eventual self-adjoint words are L^2-dense in $L^2(R)_{\mathrm{sa}}$.

The right-hand side below is closed, since $L^2(C)_{\mathrm{sa}}$ is orthogonal to $\overline{\mathrm{span}_{\mathbb{R}}}(\Omega_0 \cup U)$. This orthogonality follows from $C \subset N$, $E_N(\Omega_0) = 0$, $E_N(U) = 0$, and $U \perp \Omega$. Hence

$$L^2(R)_{\mathrm{sa}} \subset L^2(C)_{\mathrm{sa}} + \overline{\mathrm{span}_{\mathbb{R}}}(\Omega_0 \cup U).$$

The reverse inclusion is clear from the definition of R. This proves the completed-block decomposition. □

7. The all-cardinal certified extension theorem

Before passing from the countable construction to arbitrary cardinals, we record the standard continuity property of nested completed blocks. It is an immediate consequence of Lemma 3.1 *(ii)* and of the continuity of orthogonal direct sums under increasing L^2-closures, but its explicit formulation will simplify all limit-stage arguments below.

Lemma 7.1. *Let $(C_i, R_i, T_i)_{i \in I}$ be an increasing chain of completed blocks, in the sense that $C_i \subset C_j$, $R_i \subset R_j$, and $T_i \subset T_j$ whenever $i \le j$. Put*

$$C = W^*\Big(\bigcup_i C_i\Big), \qquad R = W^*\Big(\bigcup_i R_i\Big), \qquad T = \bigcup_i T_i.$$

Then

$$L^2(R)_{\mathrm{sa}} = L^2(C)_{\mathrm{sa}} \oplus \overline{\mathrm{span}_{\mathbb{R}}} T.$$

Proof. In a finite von Neumann algebra, if $R = W^*(\bigcup_i R_i)$ for an increasing family, then

$$L^2(R) = \overline{\bigcup_i L^2(R_i)}^{\|\cdot\|_2}.$$

Indeed, the strong-$*$ closure of the algebraic union is R, and bounded strong-$*$ convergence implies L^2-convergence; Kaplansky density then gives the equality. The same holds for C.

The subspace $L^2(C)_{\mathrm{sa}}$ is orthogonal to $\overline{\mathrm{span}_{\mathbb{R}}} T$, because this orthogonality holds at every stage and passes to L^2-closures. The closed sum contains each $L^2(R_i)_{\mathrm{sa}}$, hence contains the L^2-closure of their union, which is $L^2(R)_{\mathrm{sa}}$. The reverse inclusion is immediate from $C \subset R$ and $T \subset R$. □

The following remark isolates a standard cardinal-arithmetic point which is easy to obscure in the transfinite bookkeeping. No regularity assumption is imposed on either the intermediate cardinal λ or the ambient density κ; in particular, the argument applies unchanged when κ is singular.

Remark 7.2. No regularity assumption on κ is used. We identify cardinals with initial ordinals. If $\lambda > \aleph_0$ is an initial cardinal and $\alpha < \lambda$, then $|\alpha| < \lambda$, and hence

$$\max\{\aleph_0, |\alpha|\} < \lambda.$$

The case $\lambda = \aleph_0$ is handled separately as the base case in Proposition 7.3; in that case the displayed strict inequality with λ on the right is not used.

In all applications below, whenever $\lambda < \kappa$ and $\alpha < \lambda$, one still has

$$\max\{\aleph_0, |\alpha|\} < \kappa.$$

Indeed, if $\lambda = \aleph_0$, then the left-hand side is $\aleph_0 < \kappa$; if $\lambda > \aleph_0$, then the preceding paragraph gives $\max\{\aleph_0, |\alpha|\} < \lambda < \kappa$. Thus the argument does not use regularity of κ, and this remains true when κ is singular.

The next proposition upgrades the countable certified extension to every cardinal below the ambient L^2-density. Its transfinite organization is reminiscent of projectional resolutions of the identity in nonseparable Banach space theory [Kub09], but the successor step here has an additional operator-algebraic difficulty: the previously constructed symmetries must themselves become part of the certified background for the next application of the induction hypothesis.

This is achieved by re-certifying the entire previously constructed family as a single layer over its old support. Consequently the finite depth of the certificate does not accumulate along the transfinite recursion. At limit stages the completed blocks are passed to their increasing union, while the chain-continuity of the strong closure operation preserves certification. The cardinal estimates use only that the current ordinal has cardinality strictly below the induction cardinal, and hence require no regularity assumption. This all-cardinal extension theorem is a principal new component of the proof.

Proposition 7.3 (Certified extension theorem)**.** *Let $\kappa = \operatorname{dens} L^2(M) > \aleph_0$ and let $\aleph_0 \le \lambda < \kappa$. Let $N \subset M$ satisfy $\operatorname{dens} L^2(N) < \kappa$. Let $\Omega \subset \mathcal{S}_0(M)$ be a finite-layer certified background with $|\Omega| < \kappa$, $E_N(\Omega) = 0$, and a fixed strong closure operation cl_Ω as in Proposition 5.4. Let*

$$S \subset \Omega, \qquad X \subset M_{\mathrm{sa}}, \qquad Y \subset N_{\mathrm{sa}}$$

with $|S|, |X|, |Y| \le \lambda$. Then there are von Neumann algebras $C \subset N$ and $C \subset R \subset M$, a cl_Ω-closed set $\Omega_0 \subset \Omega$, and an orthonormal family $U \subset \mathcal{S}_0(M)$ such that

$$S \subset \Omega_0, \qquad X \subset R, \qquad Y \subset C,$$
$$|\Omega_0|,\ |U|,\ \operatorname{dens} L^2(R) \le \lambda,$$
$$E_N(U) = 0, \qquad U \perp \Omega,$$

and

$$L^2(R)_{\mathrm{sa}} = L^2(C)_{\mathrm{sa}} \oplus \overline{\operatorname{span}_{\mathbb{R}}}(\Omega_0 \cup U).$$

Thus U is a layer over Ω_0.

Proof. We prove the proposition by induction on λ. The case $\lambda = \aleph_0$ is Lemma 6.3.

Assume $\lambda > \aleph_0$ and that the proposition has been proved for all cardinals μ with $\aleph_0 \le \mu < \lambda$. Enumerate the disjoint union of the data S, X, and Y as $(z_\alpha)_{\alpha<\lambda}$, with repetitions or dummy entries if necessary, remembering the type of each genuine datum. We first apply the countable case to the original background Ω, with support $\mathrm{cl}_\Omega(\varnothing)$ and with $X = Y = \varnothing$. This gives an initial completed state

$$(C_0, R_0, \Omega_0 \cup U_0)$$

with countable density, where Ω_0 is cl_Ω-closed and U_0 is a layer over Ω_0. The recursion below starts from this state; at successor stages we enlarge the current state to absorb the next datum.

We recursively construct, for $\alpha < \lambda$, completed states

$$(C_\alpha, R_\alpha, \Omega_\alpha \cup U_\alpha)$$

such that

$$C_\alpha \subset N, \quad C_\alpha \subset R_\alpha \subset M, \quad \Omega_\alpha \subset \Omega, \quad \Omega_\alpha = \mathrm{cl}_\Omega(\Omega_\alpha),$$

U_α is a layer over Ω_α,

$$E_N(U_\alpha) = 0, \quad U_\alpha \perp \Omega,$$

and

$$\operatorname{dens} L^2(R_\alpha),\ |\Omega_\alpha|,\ |U_\alpha| \le \max\{\aleph_0, |\alpha|\}.$$

The states are required to be increasing in all coordinates.

Suppose the state at α is constructed. Put

$$\lambda_\alpha = \max\{\aleph_0, |\alpha|\} < \lambda.$$

Here λ is identified with its initial ordinal. No regularity of λ, and hence no regularity of κ, is used in this inequality; see Remark 7.2. By the induction invariant,

$$\Omega_\alpha = \mathrm{cl}_\Omega(\Omega_\alpha)$$

and

$$L^2(R_\alpha)_{\mathrm{sa}} = L^2(C_\alpha)_{\mathrm{sa}} \oplus \overline{\mathrm{span}_{\mathbb{R}}}(\Omega_\alpha \cup U_\alpha).$$

Moreover U_α is orthonormal and orthogonal to Ω. Therefore Corollary 5.5, applied with $S = \Omega_\alpha$ and $U = U_\alpha$, certifies the background

$$\Omega \dot\cup U_\alpha.$$

This is a re-certification of the whole currently constructed family U_α as a single layer over Ω_α; the finite certificate depth is not accumulated along the transfinite recursion. The same corollary supplies a strong closure operation on $\Omega\dot\cup U_\alpha$ compatible with the original operation cl_Ω. Also

$$E_N(\Omega\dot\cup U_\alpha) = 0$$

and

$$|\Omega\dot\cup U_\alpha| < \kappa,$$

because $|\Omega| < \kappa$ and $|U_\alpha| \le \lambda_\alpha < \lambda < \kappa$.

Choose self-adjoint L^2-dense sets

$$Y_\alpha \subset (C_\alpha)_{\mathrm{sa}}, \qquad X_\alpha \subset (R_\alpha)_{\mathrm{sa}},$$

of cardinality at most λ_α. This is possible because

$$\operatorname{dens} L^2(C_\alpha) \le \operatorname{dens} L^2(R_\alpha) \le \lambda_\alpha$$

and the bounded self-adjoint parts are L^2-dense in the corresponding self-adjoint L^2-spaces. By Lemma 3.1(iv),

$$W^*(Y_\alpha) = C_\alpha, \qquad W^*(X_\alpha) = R_\alpha.$$

Set

$$S'_\alpha = \Omega_\alpha \cup U_\alpha, \qquad X'_\alpha = X_\alpha, \qquad Y'_\alpha = Y_\alpha.$$

If the datum z_α comes from the original set S, add it to S'_α; if it comes from X, add it to X'_α; and if it comes from Y, add it to Y'_α. Apply the induction hypothesis at cardinal λ_α to the background $\Omega\dot\cup U_\alpha$ with input $S'_\alpha, X'_\alpha, Y'_\alpha$. We obtain a completed extension

$$(C_{\alpha+1}, R_{\alpha+1}, \Xi_{\alpha+1} \cup V_{\alpha+1}),$$

where $\Xi_{\alpha+1} \subset \Omega\dot\cup U_\alpha$ is closed for the chosen closure operation on $\Omega\dot\cup U_\alpha$ and contains $\Omega_\alpha \cup U_\alpha$. Since all of U_α is required support,

$$\Xi_{\alpha+1} = (\Xi_{\alpha+1} \cap \Omega)\dot\cup U_\alpha.$$

Set

$$\Omega_{\alpha+1} = \Xi_{\alpha+1} \cap \Omega, \qquad U_{\alpha+1} = U_\alpha \cup V_{\alpha+1}.$$

By the compatibility part of Proposition 5.4, $\Omega_{\alpha+1}$ is cl_Ω-closed. Moreover the new state contains the old one. Indeed, $Y_\alpha \subset C_{\alpha+1}$ and $X_\alpha \subset R_{\alpha+1}$ by the choice of the input data for the induction hypothesis; since

$$W^*(Y_\alpha) = C_\alpha, \qquad W^*(X_\alpha) = R_\alpha,$$

we get

$$C_\alpha \subset C_{\alpha+1}, \qquad R_\alpha \subset R_{\alpha+1}.$$

Also $\Omega_\alpha \subset \Omega_{\alpha+1}$ and $U_\alpha \subset U_{\alpha+1}$ by construction. Thus the completed states are increasing in all coordinates.

The completed-block decomposition for the output says exactly that

$$L^2(R_{\alpha+1})_{\mathrm{sa}} = L^2(C_{\alpha+1})_{\mathrm{sa}} \oplus \overline{\mathrm{span}_{\mathbb{R}}}(\Omega_{\alpha+1} \cup U_{\alpha+1}).$$

The cardinal bounds are $\leq \lambda_\alpha \leq \max\{\aleph_0, |\alpha+1|\}$. Thus the successor step is complete.

At a limit ordinal $\delta < \lambda$, define

$$C_\delta = W^*\Big(\bigcup_{\alpha<\delta} C_\alpha\Big), \qquad R_\delta = W^*\Big(\bigcup_{\alpha<\delta} R_\alpha\Big),$$

$$\Omega_\delta = \bigcup_{\alpha<\delta} \Omega_\alpha, \qquad U_\delta = \bigcup_{\alpha<\delta} U_\alpha.$$

Since $\delta < \lambda$ and λ is an initial cardinal, $|\delta| < \lambda$. The cardinal estimates are as follows. For each $\alpha < \delta$, choose an L^2-dense subset $D_\alpha \subset L^2(R_\alpha)$ of cardinal at most $\max\{\aleph_0, |\alpha|\}$. Then

$$\Big|\bigcup_{\alpha<\delta} D_\alpha\Big| \leq |\delta| \cdot \max\{\aleph_0, |\delta|\} = \max\{\aleph_0, |\delta|\}.$$

By Lemma 3.1(ii), this union is L^2-dense in $L^2(R_\delta)$. The same cardinal calculation gives

$$|\Omega_\delta|,\ |U_\delta| \leq \max\{\aleph_0, |\delta|\}.$$

Hence

$$\operatorname{dens} L^2(R_\delta),\ |\Omega_\delta|,\ |U_\delta| \leq \max\{\aleph_0, |\delta|\} < \lambda.$$

The continuity of cl_Ω from Proposition 5.4 shows that Ω_δ is cl_Ω-closed. Lemma 7.1 applied to the nested completed blocks gives

$$L^2(R_\delta)_{\mathrm{sa}} = L^2(C_\delta)_{\mathrm{sa}} \oplus \overline{\mathrm{span}_{\mathbb{R}}}(\Omega_\delta \cup U_\delta).$$

Thus U_δ is a layer over Ω_δ.

Finally, put

$$C = W^*\Big(\bigcup_{\alpha<\lambda} C_\alpha\Big), \qquad R = W^*\Big(\bigcup_{\alpha<\lambda} R_\alpha\Big),$$

$$\Omega_{\mathrm{out}} = \bigcup_{\alpha<\lambda} \Omega_\alpha, \qquad U = \bigcup_{\alpha<\lambda} U_\alpha.$$

The continuity of cl_Ω gives that Ω_{out} is cl_Ω-closed, and Lemma 7.1 gives

$$L^2(R)_{\mathrm{sa}} = L^2(C)_{\mathrm{sa}} \oplus \overline{\mathrm{span}_{\mathbb{R}}}(\Omega_{\mathrm{out}} \cup U).$$

All data from S, X, and Y were absorbed during the recursion. The cardinal bounds for Ω_{out} and U follow from taking a union of λ many sets each of cardinal at most λ:

$$|\Omega_{\mathrm{out}}|,\ |U| \leq \lambda.$$

It remains only to record the density bound for R. For each $\alpha < \lambda$, choose an L^2-dense set $D_\alpha \subset L^2(R_\alpha)$ with $|D_\alpha| \leq \lambda$. Then

$$\Big|\bigcup_{\alpha<\lambda} D_\alpha\Big| \leq \lambda \cdot \lambda = \lambda,$$

where the last equality uses the standard cardinal arithmetic identity $\lambda \cdot \lambda = \lambda$ for every infinite cardinal λ. By Lemma 3.1*(ii)*, the union $\bigcup_{\alpha<\lambda} D_\alpha$ is L^2-dense in $L^2(R)$. Hence

$$\operatorname{dens} L^2(R) \leq \lambda.$$

Thus the conclusion holds with $\Omega_0 = \Omega_{\mathrm{out}}$. ∎

8. Relative extension and proof of Theorem 1.2

We now extract from Proposition 7.3 the relative basis extension property needed in the final recursion. This formulation may be compared with the extension theorem of De and Mukherjee [DM23, Theorem 3.4], which extends uniformly bounded self-adjoint bases in a separable GNS space. Here the added vectors are required to be trace-zero symmetries, the original symmetry basis is preserved exactly, a prescribed set of new elements is absorbed, and the density of the enlarged algebra remains controlled.

The proof is formally short, because the all-cardinal certified extension theorem has already incorporated the required norming, boundedness, and bookkeeping arguments. The resulting relative symmetry-basis extension statement is the successor-step principle used in the proof of the main theorem.

Proposition 8.1 (Relative extension property)**.** *Let $\kappa = \operatorname{dens} L^2(M) > \aleph_0$ and let $\aleph_0 \le \lambda < \kappa$. Let $N \subset M$ be a von Neumann subalgebra with $\operatorname{dens} L^2(N) \le \lambda$. Suppose $H_0(N)$ has an orthonormal basis $B_N \subset \mathcal{S}_0(N)$. If $X \subset M_{\mathrm{sa}}$ and $|X| \le \lambda$, then there are a von Neumann algebra P with $N \cup X \subset P \subset M$ and an orthonormal family $U \subset \mathcal{S}_0(P)$ such that*

$$\operatorname{dens} L^2(P) \le \lambda, \qquad U \perp L^2(N)_{\mathrm{sa}},$$

and $B_N \cup U$ is an orthonormal basis of $H_0(P)$.

Proof. Apply Proposition 7.3 with the empty background $\Omega = \varnothing$, $S = \varnothing$, and with $Y \subset N_{\mathrm{sa}}$ a self-adjoint L^2-dense set of cardinal $\le \lambda$. We obtain $C \subset N$, $C \subset R \subset M$, and $U \subset \mathcal{S}_0(M)$ such that $Y \subset C$, $X \subset R$, and

$$L^2(R)_{\mathrm{sa}} = L^2(C)_{\mathrm{sa}} \oplus \overline{\operatorname{span}_{\mathbb{R}}} U.$$

Since Y is L^2-dense in $L^2(N)_{\mathrm{sa}}$, Lemma 3.1*(iv)* gives $W^*(Y) = N$. Hence $C = N$. Put $P = R$. Then $N = C \subset R = P$ and $X \subset P$, and

$$L^2(P)_{\mathrm{sa}} = L^2(N)_{\mathrm{sa}} \oplus \overline{\operatorname{span}_{\mathbb{R}}} U.$$

In particular,

$$U \perp L^2(N)_{\mathrm{sa}}.$$

Moreover $U \subset P$. Indeed, each $u \in U$ belongs to M, and its L^2-vector belongs to $L^2(P)$ by the displayed decomposition. Hence Lemma 3.1*(iii)* gives $u \in P$. Since each $u \in U$ is a trace-zero symmetry in M, it is also a trace-zero symmetry in P. Thus $U \subset \mathcal{S}_0(P)$.

Since $U \subset \mathcal{S}_0(P)$, every element of U is orthogonal to 1. Taking the orthogonal complement of $\mathbb{R}1$ in the displayed decomposition gives

$$H_0(P) = H_0(N) \oplus \overline{\operatorname{span}_{\mathbb{R}}} U.$$

Since B_N is an orthonormal basis of $H_0(N)$, the family $B_N \cup U$ is an orthonormal basis of $H_0(P)$. □

The transfinite recursion requires an initial subalgebra whose trace-zero real L^2-space already has a symmetry basis. We use the classical dyadic construction of a diffuse abelian subalgebra together with the Walsh orthonormal system [Wal23]. We include the short construction in order to keep track of the fact that every nonconstant basis element is a trace-zero self-adjoint unitary; no novelty is claimed for this starting lemma.

Lemma 8.2 (A separable abelian starting algebra)**.** *Every II_1 factor contains a separable diffuse abelian von Neumann subalgebra A_0. Moreover $H_0(A_0)$ has an orthonormal basis contained in $\mathcal{S}_0(A_0)$.*

Proof. We construct a dyadic system of projections. Start with $p_\varnothing = 1$. By the comparison and dimension theory of projections in a finite factor [KR97b, Ch. 6], every nonzero projection in M can be split into two orthogonal subprojections of equal trace. Recursively, for every finite binary string $\sigma \in \{0,1\}^{<\mathbb{N}}$, choose orthogonal projections $p_{\sigma 0}, p_{\sigma 1} \le p_\sigma$ such that

$$p_{\sigma 0} + p_{\sigma 1} = p_\sigma, \qquad \tau(p_{\sigma 0}) = \tau(p_{\sigma 1}) = \frac{1}{2}\tau(p_\sigma).$$

The projections in this dyadic tree commute pairwise, since any two of them are either orthogonal or one is dominated by the other. Let A_0 be the abelian von Neumann algebra generated by all projections p_σ. For $n \ge 0$, put

$$F_n = \operatorname{span}\{p_\sigma : |\sigma| = n\}.$$

Then $F_n \subset F_{n+1}$, and

$$A_0 = W^*\Big(\bigcup_{n\ge 0} F_n\Big).$$

Hence, by Lemma 3.1*(ii)*,

$$L^2(A_0) = \overline{\bigcup_{n\ge 0} L^2(F_n)}^{\|\cdot\|_2},$$

so $L^2(A_0)$ is separable.

We claim that A_0 is diffuse. Let $0 \ne q \in A_0$ be a projection. Choose n with $2^{-n} < \tau(q)$. Since $\sum_{|\sigma|=n} p_\sigma = 1$, there is some $\sigma \in \{0,1\}^n$ such that $qp_\sigma \ne 0$. If $qp_\sigma = q$, then $q \le p_\sigma$, which would imply

$$\tau(q) \le \tau(p_\sigma) = 2^{-n},$$

a contradiction. Thus

$$0 < qp_\sigma < q.$$

Since A_0 is abelian, qp_σ is a projection in A_0. Hence q is not minimal, and A_0 is diffuse.

For every finite set $F \subset \mathbb{N}$, choose n with $F \subset \{1,\dots,n\}$, and define

$$w_F = \sum_{\sigma\in\{0,1\}^n} (-1)^{\sum_{j\in F}\sigma_j} p_\sigma.$$

This definition is independent of the choice of n. Each w_F is a self-adjoint unitary in A_0, $w_\varnothing = 1$, and $\tau(w_F) = 0$ whenever $F \ne \varnothing$. For each n, the finite family

$$\{w_F : F \subset \{1,\dots,n\}\}$$

is the usual Walsh orthonormal basis of $L^2(F_n)_{\mathrm{sa}}$. Therefore

$$\{w_F : F \subset \mathbb{N},\ |F| < \infty\}$$

is an orthonormal basis of $L^2(A_0)_{\mathrm{sa}}$, and the nonconstant Walsh symmetries

$$\{w_F : 0 < |F| < \infty\}$$

form an orthonormal basis of $H_0(A_0)$ contained in $\mathcal{S}_0(A_0)$. □

Now, we prove our main result Theorem 1.2. The new extension machinery makes it possible to run a recursion through the full density character of $L^2(M,\tau)$, while every intermediate algebra retains strictly smaller density and an orthonormal basis of trace-zero symmetries. At each successor stage one prescribed member of an L^2-dense family is absorbed, and at limit stages the nested bases and subalgebras are passed to their unions.

Proof of Theorem 1.2. Let
$$\kappa = \operatorname{dens} L^2(M,\tau) > \aleph_0.$$
The decompositions
$$L^2(M)_{\mathrm{sa}} = H_0(M) \oplus \mathbb{R}1, \qquad L^2(M) = L^2(M)_{\mathrm{sa}} + iL^2(M)_{\mathrm{sa}}$$
show that $\operatorname{dens} H_0(M) = \kappa$. The bounded trace-zero self-adjoint elements $M_{\mathrm{sa}} \cap H_0(M)$ are L^2-dense in $H_0(M)$. Indeed, if $\xi \in H_0(M)$, then by definition of $L^2(M)_{\mathrm{sa}}$ there are $x_i = x_i^* \in M$ with $\|x_i - \xi\|_2 \to 0$. Since
$$\tau(x_i) = \langle x_i, 1\rangle_2 \longrightarrow \langle \xi, 1\rangle_2 = 0,$$
we have
$$x_i - \tau(x_i)1 \in M_{\mathrm{sa}} \cap H_0(M)$$
and
$$\|x_i - \tau(x_i)1 - \xi\|_2 \to 0.$$
Choose an L^2-dense family
$$\{a_\alpha : \alpha < \kappa\} \subset M_{\mathrm{sa}} \cap H_0(M).$$

By Lemma 8.2, choose a separable diffuse abelian von Neumann subalgebra $N_0 \subset M$ and an orthonormal basis $B_0 \subset \mathcal{S}_0(N_0)$ of $H_0(N_0)$.

We recursively construct von Neumann algebras $N_\alpha \subset M$ and orthonormal families $B_\alpha \subset \mathcal{S}_0(M)$, for $\alpha < \kappa$, such that:

(i) $B_\alpha \subset \mathcal{S}_0(N_\alpha)$ is an orthonormal basis of $H_0(N_\alpha)$;
(ii) $a_\alpha \in N_{\alpha+1}$;
(iii) if $\alpha < \beta$, then $N_\alpha \subset N_\beta$ and $B_\alpha \subset B_\beta$;
(iv) $\operatorname{dens} L^2(N_\alpha) \le \max\{\aleph_0, |\alpha|\}$.

The initial pair is (N_0, B_0).

Suppose N_α, B_α have been constructed. Put
$$\lambda_\alpha = \max\{\aleph_0, |\alpha|\}.$$
Since $\alpha < \kappa$ and κ is identified with its initial ordinal, $\lambda_\alpha < \kappa$. Proposition 8.1, applied to $N = N_\alpha$ and $X = \{a_\alpha\}$, gives a von Neumann algebra $N_{\alpha+1}$ containing $N_\alpha \cup \{a_\alpha\}$ and an orthonormal family $U_\alpha \subset \mathcal{S}_0(N_{\alpha+1})$ such that
$$B_\alpha \cup U_\alpha$$
is an orthonormal basis of $H_0(N_{\alpha+1})$. Set
$$B_{\alpha+1} = B_\alpha \cup U_\alpha.$$
The density bound is also supplied by Proposition 8.1.

At a limit ordinal $\delta < \kappa$, set
$$N_\delta = W^*\Big(\bigcup_{\alpha<\delta} N_\alpha\Big), \qquad B_\delta = \bigcup_{\alpha<\delta} B_\alpha.$$
In a finite von Neumann algebra,
$$L^2(N_\delta) = \overline{\bigcup_{\alpha<\delta} L^2(N_\alpha)}^{\|\cdot\|_2},$$
by the same Kaplansky-density argument as in Lemma 7.1. We claim that
$$H_0(N_\delta) = \overline{\bigcup_{\alpha<\delta} H_0(N_\alpha)}^{\|\cdot\|_2}.$$

Let $\xi \in H_0(N_\delta)$ and let $\varepsilon > 0$. By the displayed L^2-continuity and by taking self-adjoint parts, there are $\alpha < \delta$ and $x = x^* \in N_\alpha$ such that

$$\|x - \xi\|_2 < \varepsilon.$$

Since $\xi \perp 1$ and $\|1\|_2 = 1$,

$$|\tau(x)| = |\langle x - \xi, 1\rangle_2| \leq \|x - \xi\|_2 < \varepsilon.$$

Therefore $x - \tau(x)1 \in H_0(N_\alpha)$, and

$$\|x - \tau(x)1 - \xi\|_2 \leq \|x - \xi\|_2 + |\tau(x)|\,\|1\|_2 < 2\varepsilon.$$

This proves the claim. Since the bases are nested, B_δ is an orthonormal basis of $H_0(N_\delta)$. Also

$$\operatorname{dens} L^2(N_\delta) \leq \max\{\aleph_0, |\delta|\} < \kappa.$$

Put

$$B = \bigcup_{\alpha<\kappa} B_\alpha.$$

Since the B_α's are nested orthonormal families, B is orthonormal, and $B \subset \mathcal{S}_0(M)$. For every $\alpha < \kappa$, the element a_α belongs to $N_{\alpha+1}$ and has trace zero; hence

$$a_\alpha \in H_0(N_{\alpha+1}) \subset \overline{\operatorname{span}_{\mathbb{R}}} B.$$

Since the family $(a_\alpha)_{\alpha<\kappa}$ is L^2-dense in $H_0(M)$, the orthonormal family B is a Hilbert orthonormal basis of $H_0(M)$. Hence $|B| = \operatorname{dens} H_0(M) = \kappa$, because the density character of an infinite Hilbert space equals the cardinality of any Hilbert orthonormal basis. By Lemma 2.2, $\{\widehat{1}\} \cup \{\widehat{s} : s \in B\}$ is a Hilbert orthonormal basis of $L^2(M, \tau)$ consisting of self-adjoint unitaries. Since $\kappa > \aleph_0$, this basis has cardinality κ. Each of these vectors is a trace vector, again by Lemma 2.2. $\square$

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Appendix A. Corners and finite amplification

The following full-corner density identity is standard in spirit, although we include a proof in order to make the cardinal bookkeeping explicit. A nonzero corner of a type II_1 factor is a full corner and is therefore W^*-Morita equivalent to the original factor in the sense of Rieffel [Rie74]. In the finite-factor setting this equivalence can be realized concretely by a finite amplification

$$M \cong M_n(fMf), \qquad f \leq e,$$

and finite Hilbert direct sums do not alter density character. We use the standard continuous dimension and comparison theory of projections [KR97b, Chapters 6 and 8], together with the terminology for cardinal invariants used in [She12].

Lemma A.1. *Let M be a II_1 factor with faithful normal tracial state τ, and let $0 \neq e \in M$ be a projection. Equip eMe with its normalized trace $\tau_e(x) = \tau(x)/\tau(e)$, where $x \in eMe$. Then*

$$\operatorname{dens} L^2(eMe, \tau_e) = \operatorname{dens} L^2(M, \tau).$$

In particular, if $\operatorname{dens} L^2(M, \tau) = \kappa > \aleph_0$, then

$$\operatorname{dens} L^2(eMe, \tau_e) = \kappa.$$

Proof. Up to multiplication of the norm by the constant $\tau(e)^{-1/2}$, the space $L^2(eMe, \tau_e)$ identifies with the closed subspace $eL^2(M, \tau)e$ of $L^2(M, \tau)$. Consequently,

$$\operatorname{dens} L^2(eMe, \tau_e) \leq \operatorname{dens} L^2(M, \tau).$$

For the reverse inequality, choose $n \in \mathbb{N}$ such that $1/n \leq \tau(e)$. By the continuous dimension theory of II_1 factors, there exist mutually orthogonal projections $e_1, \ldots, e_n \in M$ and a projection $f \leq e$ such that

$$\sum_{i=1}^{n} e_i = 1, \qquad \tau(e_i) = \tau(f) = \frac{1}{n} \quad (1 \leq i \leq n).$$

Since projections of equal trace in a finite factor are Murray–von Neumann equivalent, we may choose partial isometries $v_1, \dots, v_n \in M$ satisfying

$$v_i^* v_i = f, \qquad v_i v_i^* = e_i.$$

Define

$$\Phi : M \longrightarrow M_n(fMf), \qquad \Phi(x) = \big(v_i^* x v_j\big)_{i,j=1}^n.$$

Since $\sum_i e_i = 1$, a direct computation shows that Φ is a unital $*$-homomorphism. Its inverse is given by

$$\Phi^{-1}\big((a_{ij})_{i,j}\big) = \sum_{i,j=1}^n v_i a_{ij} v_j^*, \qquad (a_{ij})_{i,j} \in M_n(fMf).$$

Hence

$$M \cong M_n(fMf).$$

Let

$$\tau_f = \frac{1}{\tau(f)}\,\tau|_{fMf} = n\tau|_{fMf},$$

and let tr_n denote the normalized trace on $M_n(\mathbb{C})$. For every $x \in M$,

$$\begin{aligned}(\mathrm{tr}_n \otimes \tau_f)(\Phi(x)) &= \frac{1}{n}\sum_{i=1}^n \tau_f(v_i^* x v_i)\\ &= \sum_{i=1}^n \tau(v_i^* x v_i)\\ &= \sum_{i=1}^n \tau(x e_i) = \tau(x).\end{aligned}$$

Thus Φ is trace preserving and extends to a unitary

$$L^2(M,\tau) \cong L^2\big(M_n(fMf), \mathrm{tr}_n \otimes \tau_f\big).$$

As a Hilbert space, $L^2(M_n(fMf))$ is a finite direct sum of n^2 copies of $L^2(fMf, \tau_f)$. Since fMf is again a II_1 factor, it follows that

$$\operatorname{dens} L^2(M,\tau) = \operatorname{dens} L^2(fMf, \tau_f).$$

Finally, after a constant rescaling of the norm, $L^2(fMf, \tau_f)$ identifies with the closed subspace

$$fL^2(eMe, \tau_e)f \subset L^2(eMe, \tau_e).$$

Therefore

$$\operatorname{dens} L^2(M,\tau) = \operatorname{dens} L^2(fMf,\tau_f) \le \operatorname{dens} L^2(eMe, \tau_e).$$

Combining the two inequalities proves the assertion. □

Remark A.2. Since M is a factor, every nonzero projection in M has central support 1. Thus eMe is a full corner of M, and the M-eMe bimodule Me implements a W^*-Morita equivalence in the sense of [Rie74].

In the present finite setting, the condition $\tau(e) > 0$ implies that finitely many copies of e dominate 1 in the Murray–von Neumann order. The isomorphism

$$M \cong M_n(fMf), \qquad f \le e,$$

appearing in the proof is a concrete finite-amplification realization of this equivalence. Since finite Hilbert direct sums do not change density character, the lemma may be viewed as the corresponding L^2-density consequence of the full-corner relation.

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